

Research paper

Investigating bearing and gear vibrations with a Micro-Electro-Mechanical Systems (MEMS) and machine learning approach

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ABSTRACT

Bearings and gears are the pivotal components of mechanical systems and are prone to faults that can impact the system's overall performance. These components' condition monitoring and fault diagnosis are vital for maintaining system reliability and efficiency. In this research, a MEMS setup is initially developed, comprising a Raspberry Pi 4B+ CPU module, a NucleoF401RET6 MCU, an OLED screen, and an Adxl1002z accelerometer for acquiring vibration signals at the desired sampling frequency stored in the CPU memory. Further, an RF model is also developed to classify different types of faults based on features extracted from the acquired vibration data. The model evaluates the precision and reliability of the MEMS setup in capturing and classifying vibration signals. A detailed signal analysis is also conducted to determine the performance of the developed MEMS setup and to investigate the effect of bearing vibration signature due to gear fault and vice versa. The results indicate that bearing faults cause irregularities in the shaft's rotational speed, leading to modulation of the gear mesh frequency (gmf) of gears mounted on the affected shaft. Conversely, gear faults disrupt the shaft's rotational motion, imposing excessive loads on shaft-supported bearings. These disruptions result in distinct vibration patterns characterised by increased harmonics and side bands within the bearing frequency range. The RF model effectively identifies and classifies faults with high accuracy by leveraging its ability to prioritise the most significant vibrational features, resulting in improved predictive performance and robustness.

List of symbols

B_d - Diameter of the ball	f_{CF} - Centre Frequency
D_i - Diameter of the inner raceway	f_{MF} - Mean Frequency
D_o - Diameter of the outer raceway	f_{RMSF} - Root Mean Square Frequency
F_S - Sampling Frequency	f_{RVF} - Root Variance Frequency
G_r - Gear ratio	f_g - Rotating frequency of the gear
N_b - Number of balls in the rolling bearing	f_p - Rotating frequency of the pinion
P_d - Pitch Diameter	f_r - Shaft rotation frequency
T_g - Number of teeth on the gear	$\bar{x}$ - Mean value
T_p - Number of teeth on the pinion	x_i - Acceleration value for the i^{th} point
T_s - Sampling Time	x_{rms} - Root Mean Square value
T_{SD} - Data Storage Delay	α - Contact angle of the ball
T_{SPI} - SPI Communication Delay	σ - Standard Deviation
T_{UART} - UART Bridge Delay	σ^2 - Variance

List of abbreviations

ADC - Analog to Digital Converter	ML - Machine Learning
CBM - Condition-Based Monitoring	CPU - Central Processing Unit
CF - Crest Factor	OLED - Organic Light Emitting Diode
CS - Chip Select	RF - Random Forest
DFT - Discrete Fourier Transform	SCL/SCK - Serial Clock
EDDM - Electric Discharge Drilling Machine	SDA - Serial Data Pin
FFT - Fast Fourier Transform	SD Card - Secure Digital Card
FTF - Fundamental Train Frequency	SF - Shape Factor
GMF - Gear Mesh Frequency	SPI - Serial Peripheral Interface
GPIO - General Purpose Input/Output	SPs - Statistical Parameters
HDMI - High-Definition Multimedia Interface	VFD - Variable Frequency Drive
HSS - High-Speed Steel	$bpfo$ - Ball Pass Frequency Outer
I ² C - Inter-Integrated Controller	$bpfi$ - Ball Pass Frequency Inner
MCU - Microcontroller Unit	gmf - Gear Mesh Frequency
MEMS - Micro Electro-mechanical System	

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1. Introduction

The reliability and functioning of the gearbox depend upon its major components, such as bearings and gears. Bearings and gears are identified as primary sources of faults within gearbox systems, impacting overall system reliability. The fault in these components occurs due to harsh operating conditions, contamination in lubrication, fluctuation/variation of operating parameters, i.e. load and speeds, improper assembly & mounting, and corrosion. During the system's working, due to these causes, the fault occurred at the outer and inner raceway of the bearing, and the tooth of the gear faces cracks and wear [1]. The prior identification of these faults is crucial for unnecessary downtime, saving money and time, and even for the operator's safety. Researchers widely adopt the vibration signal analysis approach to detect prior faults in these components. During working operation, these components produce specific vibrations. The system's condition can be continuously monitored and assessed by analysing these vibration patterns. Therefore, this study explores fault diagnosis and condition monitoring strategies, focusing on the critical role of vibration analysis in assessing machinery health.

Fig. 1 gives the schematic view of the vibration signal analysis approach. The vibration signal analysis approach involves systematic data acquisition, signal analysis, and failure pattern recognition. In the data acquisition phase, sensors capture vibration signals from machinery components. Further, signal analysis approaches such as the time domain and frequency domain are used to determine the strength of the signal. The last crucial step is failure pattern recognition, in which the deviations or anomalies of the frequency spectrum are identified to recognise the potential location of the faults using the Fast Fourier Transform (FFT) or Envelope Spectrum approaches. FFT is an algorithm that efficiently computes the Discrete Fourier Transform (DFT), enabling the conversion of a signal from its time domain to its frequency domain, which is essential for analysing the frequency components of signals. This process reveals the amplitude and phase of each frequency component, enabling a detailed analysis of the signal's spectral characteristics. FFT analysis helps to recognise the characteristic frequencies by decomposing a time-domain signal into its frequency components, creating a spectrum that highlights dominant frequencies. This enables the detection of specific frequencies associated with normal or abnormal conditions for effective fault diagnosis. This comprehensive approach enables the early detection and diagnosis of mechanical issues in rotating machinery, facilitating proactive maintenance and minimising downtime by addressing emerging failures before they escalate to the system's failure [2–4].

Nowadays, researchers widely adopt ML approaches for fault classification and detection. These approaches generally rely on advanced signal analysis, which involves gathering data for signal processing and extracting relevant features. Various classifiers, such as support vector machines (SVM), artificial neural networks (ANN), and k-nearest neighbors (kNN), have been utilised for fault classification [5]. Soomro et al. [6] reviewed that the rolling bearings are critical in machinery, and bearing failure causes 45–50 % of system failures. This study reviews bearing fault classification techniques, highlighting challenges like data imbalance, limited datasets, and evolving IoT-ML solutions. Yang et al. [7] introduced a method utilising the random forest (RF) algorithm, demonstrating high accuracy in machine fault diagnosis while maintaining fast execution times. This approach outperforms other classifiers such as kNN, SVM, and decision trees.

ML ensemble classifiers, including RF and XGBoost, have shown considerable potential in improving accuracy for bearing fault diagnosis [8,9]. Kaya et al. [10] proposed a hybrid MM-1D-LBP+1D-CNN-LSTM approach for bearing fault diagnosis, accurately identifying failure types and defect sizes and enhancing early detection capabilities. Deveci et al. [11] proposed MEMS accelerometers and hall sensor-based systems to identify electrical and mechanical faults in induction motors. The exceptional classification accuracy across various datasets is achieved, with notable fault detection and load-independent performance results. Kaya et al. [12] proposed a fault size diagnosis method for bearings using CWT time-frequency images and deep transfer learning, achieving high accuracy (96.67 %–100 %) across various datasets. Akcan et al. [13] studied the uses of entropy-based feature extraction and extreme learning machines (ELM) for diagnosing bearing faults. It demonstrates superior accuracy in detecting fault location, size, and speed, outperforming existing methods with successful results across various real-world datasets. Random forest (RF) is particularly effective for fault diagnosis problems, detecting faults by creating multiple decision trees from random subsets of training data. For each tree, it selects a random subset of features at each node to ensure diversity. Each tree independently classifies data as normal or faulty, and the result is obtained by collecting the individual tree estimations using a majority vote system. This ensures that the overall classification decision is based on the consensus of the trees within the random forest. Sensor data is pre-processed in bearing and gear fault detection, and features such as time and frequency domain features are extracted. The RF model is trained on this labelled data, learning patterns that distinguish between healthy and faulty conditions, ensuring robust and accurate fault detection [14].

In the 1980s, MEMS was first used to designate newly developed

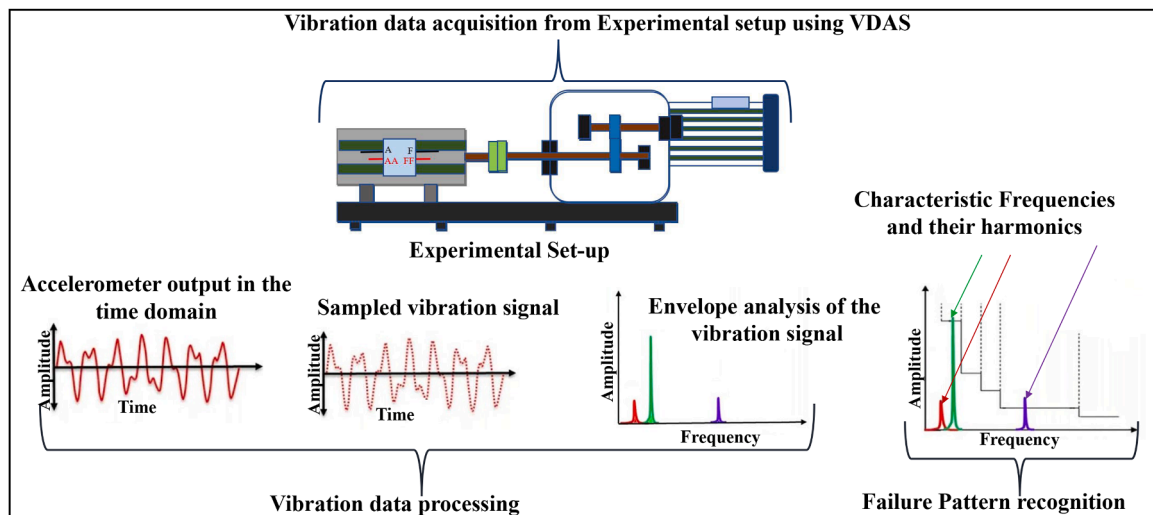


Fig. 1. Schematic view of vibration analysis approach.

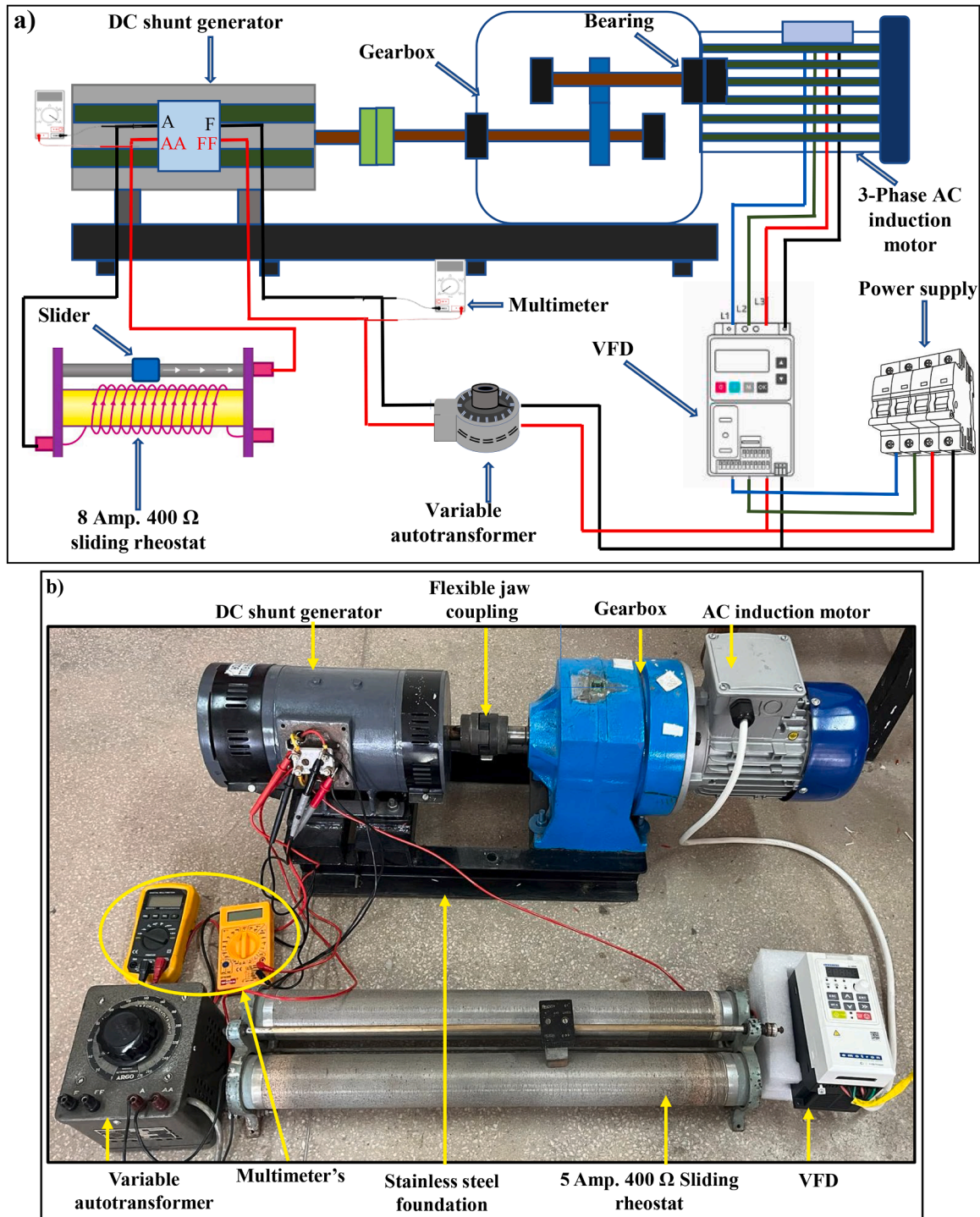


Fig. 2. (a) Schematic view of the experimental setup and (b) Actual view of an experimental setup.

mechanical systems on semiconductor chips, such as microelectronic actuators and resonators. In practice, MEMS refers to any batch-manufactured micromechanical device with a mechanical and electronic function [15,16]. In this research article, a MEMS setup is developed to capture vibration data from an experimental setup. The MEMS setup comprises Raspberry Pi 4B+, NucleoF401RET6 MCU, Adxl1002z accelerometer, and OLED screen. With its miniaturised and integrated components, the MEMS technology offers a compact and lightweight alternative to conventional sensing systems. Using a Raspberry Pi 4B+ module as the central processing unit ensures computational capabilities, connectivity, and flexibility in data handling. The

NucleoF401RET6 MCU continuously acquires and converts the analog signals from the accelerometer into digital signals for further processing and analysis. At the core of this system, an Adxl1002z accelerometer is used to capture the vibration signal. This high-performance Adxl1002z MEMS sensor is designed to accurately measure and capture vibrations across various frequencies. Its integration within the data acquisition system ensures the fidelity of acquired data, allowing for precise analysis of structural vibrations in diverse applications such as machinery monitoring [17]. Failure pattern recognition is the final phase of vibration signal analysis. This phase scrutinises the distinctive vibration signatures of bearings subjected to gear faults. Additionally, it unravels

Table 1

Specification MEMS components.

Component Name	Microprocessor / Resolution (Bits)	Maximum Clock Frequency (Hz)/Bandwidth (Hz)/Sensitivity (mV/g)/ Maximum Range (g)	Number of I/O and Analog Pins/Channels	Price	Application
Raspberry Pi 4B+ (CPU)	Quad-core Cortex-A72 (ARM v8) 64-bit	2.3 GHz	40	≈ 170 USD	Industrial embedded systems, Automotive, and IIoT
Adxl1002z	16 Bits	21 kHz/45 mV/g/ 50 g	6	≈ 100 USD	Smart industrial devices, PLC, Smart medical devices, Inbuilt Wi-Fi protocol
NucleoF401RET6 MCU	STM32F4 ARM® Cortex®-M, 12 Bit	84 MHz	56	≈ 40 USD	Industrial embedded systems, Automotive, and IIoT

the reciprocal impact of bearing faults on the vibration characteristics of gears.

Ompusunggu et al. [18] compared the results of the standard accelerometer (PCB355B12) with the MEMS-based Adxl1002 and Adxl1001 accelerometer for fault detection. The same results are obtained using the standard accelerometer and the low-cost Adxl1002 MEMS accelerometer, which are used for condition monitoring purposes. To classify tool wear using vibration analysis, Bahador et al. [19] compared the results from the Adxl1002 accelerometer with the Dytran IEPE accelerometer. It is discovered that the MEMS accelerometer's results are similar to those of the conventional accelerometer. Ref. [18, 19] used an Adxl1002z MEMS accelerometer and determined their performance with the high-end accelerometer. As a result of their findings, the Adxl1002z accelerometer was employed in this investigation to collect data. Lou et al. [20] used Raspberry Pi 3B+ CPU for data acquisition and edge computing. This research suggested an edge computing-based data-driven adaptive sampling strategy using the Modbus protocol. Battulgha et al. [21] developed a cost-effective fault detection and classification method for induction motor systems using a wireless MEMS sensor unit, offering superior sensitivity and reliability compared to conventional methods. Kumar et al. [22] developed a fault identification model using Raspberry Pi 3 B+ CPU for industrial applications. In the proposed fault identification model, Pi is an edge computing device that pre-processes vibration data.

Similarly, the Raspberry Pi is used by Refs. [20,22] to develop a fault identification model and vibration data acquisition purposes because of its unique characteristics such as in-built Wi-Fi and Bluetooth, high computing performance, low power consumption, and cost, ease of use, pre-built tools, and many built-in capabilities. Due to the advantages of Pi, the Raspberry Pi 4B+ model is used to develop a MEMS setup for this research. Mohammed et al. [23], Kafeel et al. [24], Bachar et al. [25], Mohammed et al. [26], Feng and Zuo [27], and Stander et al. [28] investigate the fault in gears using the vibration signal analysis approach. Vasić et al. [1], Dalpiaz et al. [29], Kim et al. [30], Delgado-Arredondo et al. [31], Dolenc et al. [32], Melnemy and Dai [33], Maruthi and Hegde [34] considered the bearing fault and conducted the vibration signal analysis approaches to investigate the failure pattern. The current state of research in machinery health monitoring primarily concentrates on either bearing faults or gear faults individually, resulting in a substantial gap in recognising the mutual interaction between these two components. Therefore, the present research attempts to overcome this gap by exploring the reciprocal influence of bearing faults on gear vibration signatures and vice versa. Existing studies often treat bearings and gears as isolated entities, overlooking potential synergies arising from faults in one component affecting the vibration characteristics of the other.

In this research, a MEMS setup is initially developed, comprising a Raspberry Pi 4B+ CPU module, a NucleoF401RET6 MCU, an OLED screen, and an Adxl1002z accelerometer for acquiring vibration signals. This MEMS setup ensures that vibration data is captured at the desired sampling frequency and stored in the CPU memory, effectively addressing significant limitations observed in previous studies regarding sampling frequency and data storage in MEMS systems. Further, a machine learning model is also developed to classify different types of

faults based on features extracted from the acquired vibration data. The model evaluates the precision and reliability of the MEMS setup in capturing and classifying vibration signals. Key performance metrics such as accuracy, precision, recall, and F1 score are analysed to validate the effectiveness of the developed MEMS setup in detecting various faults in the gearbox. Finally, a detailed signal analysis is conducted to evaluate the performance of the developed MEMS setup and examine the impact of bearing vibration signatures caused by gear faults and vice versa. The research aims to understand the dynamic interaction between bearing and gear vibrations, improving fault detection accuracy and prognosis in rotating machinery. A detailed explanation of the experimental setup is described in Section 2. Section 3 describes the experimental procedure. Experimental outcomes and conclusions are described in Sections 4 & 5, respectively.

2. Experimental setup

The experimental setup shown in Fig. 2(a) & (b) consists of a 3-phase induction motor connected to a gearbox that works as the prime mover. This gearbox is further connected to a DC shunt generator using a flexible jaw coupling to provide variable loading conditions, and VFD controls the speed of the prime mover. The entire assembly, including the motor, gearbox, and generator, is securely mounted on a High-Strength Steel (HSS) foundation to ensure measurement stability and accuracy. The foundational support guarantees minimal interference from external vibrations, ensuring the fidelity of the acquired vibration data. The HSS foundation is fixed to the ground to overcome the noise. The comprehensive experimental setup provides a realistic and controlled environment for investigating the reciprocal effects of bearing and gear faults on vibration signatures in rotating machinery.

In the developed experimental setup shown in Fig. 2(a) & (b), when the prime mover provides a 50 Hz (3000 rpm) input speed to the gearbox, the gearbox steps down the input speed to 400 rpm which is the input speed to the generator. An external 220 V DC input power supply is used to excite the generator's field winding, and 400 ohms of rheostat resistance is connected to the armature winding of the DC generator. The voltage produced by the generator at 3 A current is 100 V. Therefore, the generator applies 7.16 N-m of torque to the gearbox's output shaft. The torque can be varied by controlling the external DC power supply using a variable autotransformer from 0 to 220 V and by changing the resistance from 0 to 400 ohms with the help of a rheostat.

3. Experimental investigation

This section comprehensively explains the experimental procedure for acquiring and analysing the vibration data.

3.1. Component specifications of MEMS setup

A Micro-Electro-Mechanical Systems (MEMS) based data capturing system has been ingeniously designed to capture vibration signals, utilising cutting-edge technology components. This sophisticated system incorporates a Raspberry Pi 4B+ module, a NucleoF401RET6 MCU, and an Adxl1002z accelerometer. Table 1 describes the specifications of

Table 2
Configuration of the MEMS components with the CPU.

Component	Configuration
Raspberry Pi 4B+ (CPU)	Enable the CPU's SPI/I2C pins and necessary GPIOs.
Adxl1002z Accelerometer	Connect the analog (A_0), power (V_{DD}), and ground pins (GND) of the accelerometer to the same pins of the NucleoF401RET6 MCU.
NucleoF401RET6 MCU	Connect MOSI, MISO, SCK, GND, and CS pins of the NucleoF401RET6 MCU for SPI communication with the CPU.
OLED Screen	Connect the OLED screen to the CPU using a Micro HDMI cable.
Power Supply	Connect the power supply to the CPU board using a 3A, 5 V USB-C Supply.

MEMS components. Table 2 describes the configuration of the MEMS components with the CPU in which an Adxl1002z accelerometer is seamlessly interfaced with the NucleoF401RET6 MCU through analog pins, establishing a crucial link in the signal acquisition process. The NucleoF401RET6 MCU continuously acquires and converts the analog signals from the accelerometer into digital signals for further processing and analysis. The NucleoF401RET6 MCU is connected to the Raspberry Pi using the SPI protocol to facilitate continuous communication. This MEMS setup ensures precision and reliability in capturing valuable vibration data, opening up possibilities for various applications, from condition monitoring to diagnostics of industrial machinery. Fig. 3(b) describes the actual circuit diagram of the MEMS setup according to Table 2.

3.2. Functioning of MEMS setup

The complete functioning of the MEMS setup is illustrated in Fig. 3

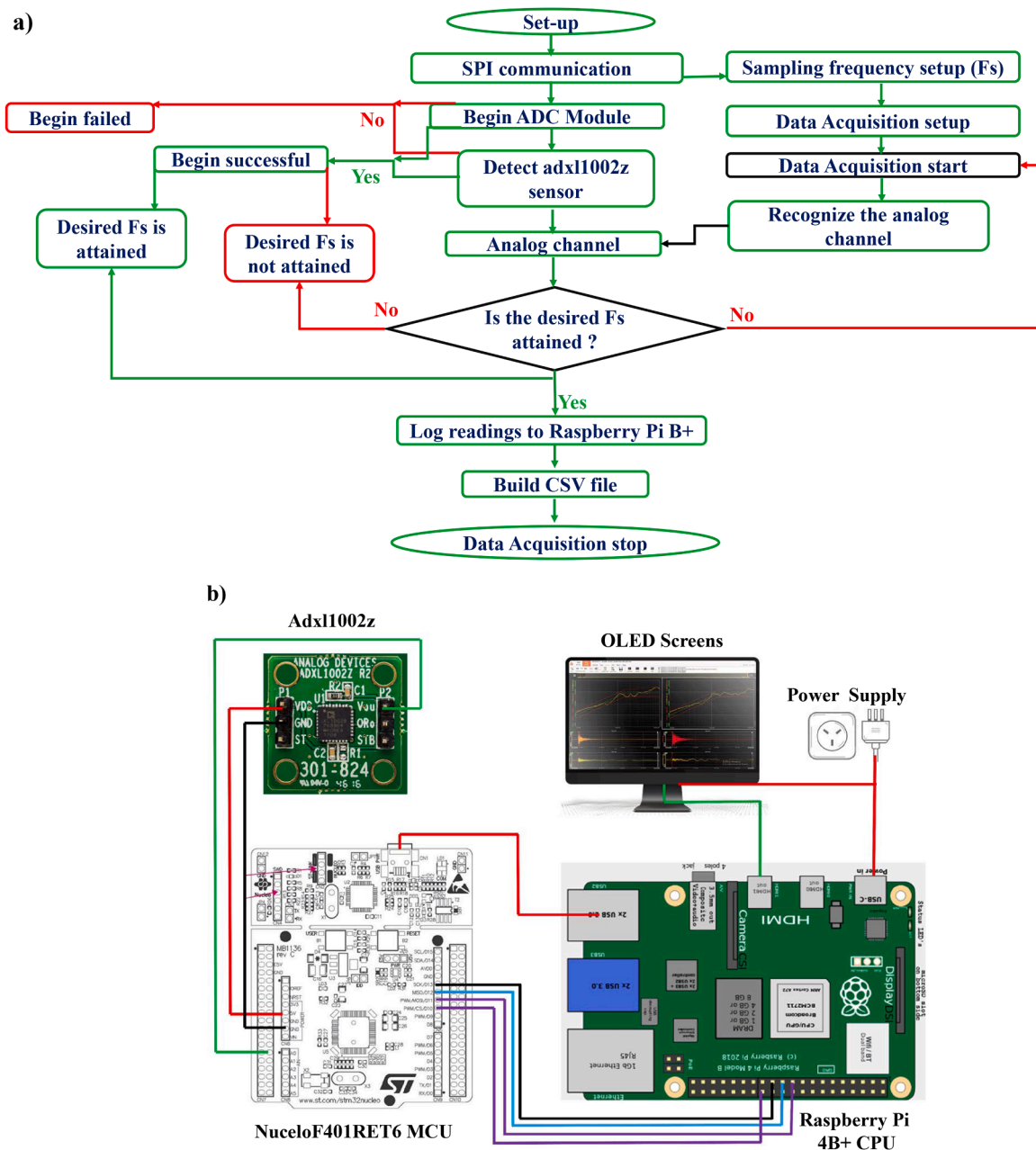


Fig. 3. (a) Functioning of MEMS Setup, and (b) Circuit diagram of MEMS setup.

Table 3

Parameters for experimentation.

S. No.	Input rpm	Variable load	Gearbox Configuration
1.	5–50 Hz (Step Size 5 Hz)	0–7.16 Nm (Step Size 1 N-m)	Healthy
2.	5–50 Hz (Step Size 5 Hz)	0–7.16 Nm (Step Size 1 N-m)	Inner Raceway Fault with (1mm, 2 mm, and 3 mm holes)
3.	5–50 Hz (Step Size 5 Hz)	0–7.16 Nm (Step Size 1 N-m)	Outer Raceway Fault (1mm, 2 mm, and 3 mm holes)
4.	5–50 Hz (Step Size 5 Hz)	0–7.16 Nm (Step Size 1 N-m)	Gear Tooth fault (30 %, 50 % and 100 % damage)

(a). According to the algorithm described in Fig. 3(a), upon powering the NucleoF401RET6 MCU with a 5 V input supply, the pre-uploaded codes, written in C++, are activated. The setup functions are invoked at the start of the code. Programming of the NucleoF401RET6 MCU is accomplished using STM32CubeIDE software developed by

STMicroelectronics. A Raspberry Pi CPU is interfaced with the NucleoF401RET6 MCU via the SPI protocol to log accelerometer readings in CSV format. Python IDLE is used to develop the code for the CPU. An OLED screen is connected to the CPU using a micro HDMI cable. Upon successful CPU initialisation, a confirmation message is displayed on the OLED screen, indicating that the setup is ready for operation. The MEMS accelerometer senses the vibration signatures, and the data is sampled using the NucleoF401RET6 MCU at the desired sampling frequency. The sampling frequency is established based on Nyquist's theorem, which dictates that it should be at least 2.56 times greater than the highest frequency in the signal to prevent aliasing. For the developed experimental setup, the maximum frequency (F_{Max}) is 544 Hz, produced by the meshed gears in gear mesh frequency (gmf). According to Nyquist's theorem, the minimum required sampling frequency is described in Eqs. (1) and (2).

$$\text{Nyquist Theorem : } F_s > 2.56 F_{Max} \quad (1)$$

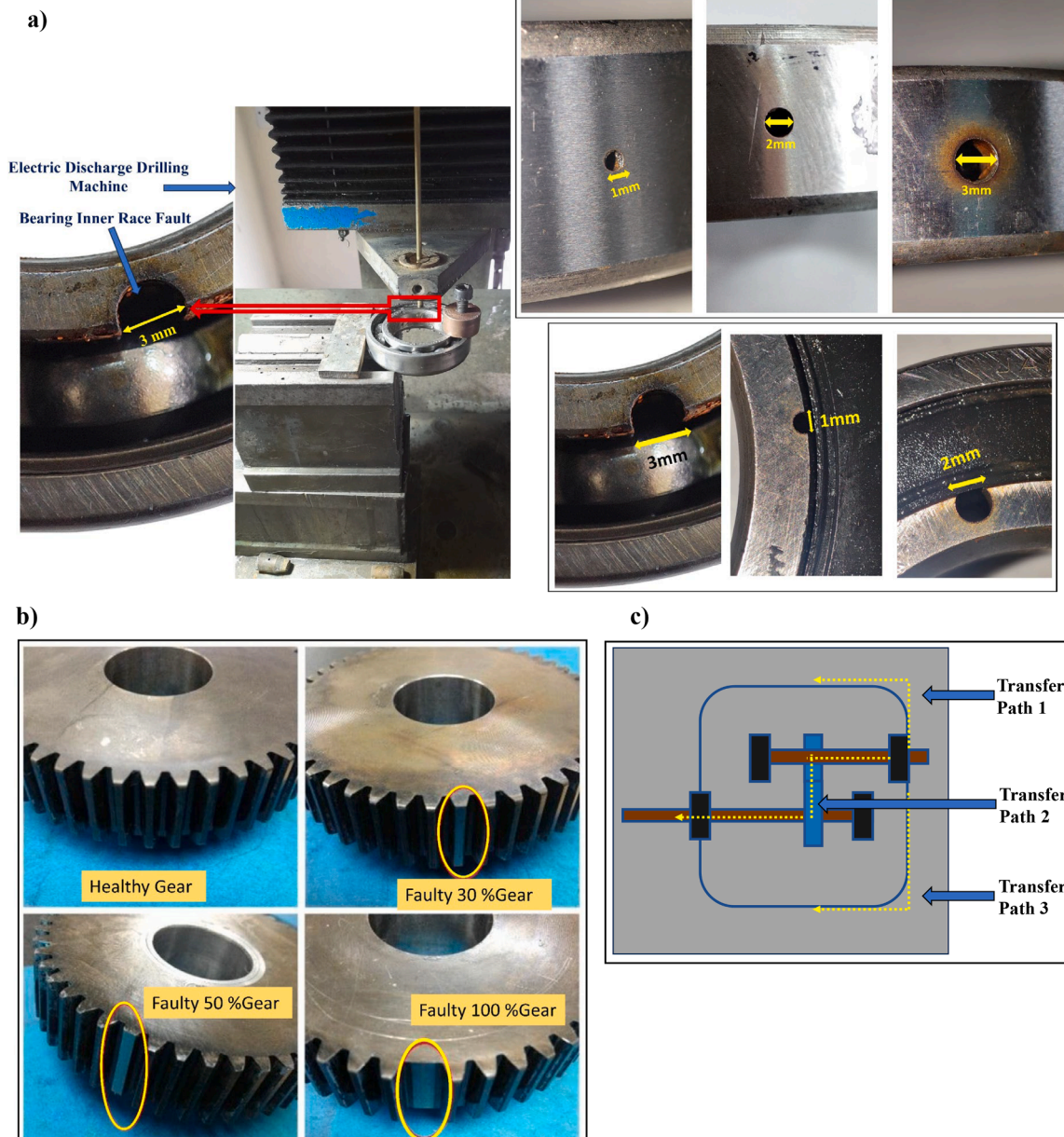


Fig. 4. Artificially induced (a) bearing inner and outer raceway fault (b) gear tooth fault (c) Vibrational signals transfer path's.

$$F_s > 2.56 * 544 = 1392 \text{ Hz} \quad (2)$$

To ensure a more accurate acquisition of the vibration data and to account for any potential higher frequency components, a sampling frequency (F_s) of 5 kHz is used in this study. In practice, a higher factor is often used for more accurate representation. This choice exceeds the minimum requirement, providing a buffer that helps in accurately capturing the signal and preventing any aliasing effects. It is observed that the NucleoF401RET6 MCU of the MEMS setup also pre-processes the vibration signals. This affects the stability of the sampling frequency. Ideally, the F_s should match the sampling period set for the microcontroller unit (NucleoF401RET6 MCU), represented as T_s . However, due to the delays caused by signal transmission (through both SPI and UART) and the time needed to store data on the micro-SD card of the CPU, the actual sampling time tends to be longer than T_s . The actual sampling frequency (denoted F_s) is modelled using Eq. (3), where F_s is the actual sampling frequency, T_s is the sampling time, T_{SPI} is the SPI communication delay, T_{UART} is the UART bridge delay, and T_{SD} refers to the data storage delay. Due to the high frequency of the internal clock, T_{SD} and T_{SPI} are negligible in Eq. (3). The UART transmission protocol is not used in the developed MEMS setup, so T_{UART} is also zero.

$$F_s = \frac{1}{T_s + T_{SPI} + T_{UART} + T_{SD}} \quad (3)$$

The acquired data at a 5 kHz sampling frequency is directly transferred and stored in the Raspberry Pi using the SPI communication protocol. The CPU is connected to an OLED screen using a micro-HDMI cable to display the results of the MEMS setup. If the system cannot attain the desired sampling frequency or fails to initialise the system, then an error message is displayed on the OLED screen to warn the user.

3.3. Experimentation

This section outlines the procedure for conducting experiments to gather vibration data for different system configurations, utilising the MEMS described in the previous section. Experiments are conducted to acquire the vibration data for healthy, faulty bearing & faulty gear configurations under various input speed and loading conditions, as shown in Table 3. In the faulty configurations, the faults are artificially induced in the system components to simulate different levels of degradation and failure. This process helps to understand the system's behaviour and response under various conditions. In the faulty bearing configuration, faults are introduced using an electric discharge drilling machine (EDDM), creating 1 mm, 2 mm, and 3 mm holes at the inner and outer raceways shown in Fig. 4(a). These faults represent varying degrees of damage: 1 mm for minor degradation, 2 mm for moderate, and 3 mm for severe failure. In the case of faulty gear configuration, faults are created by removing material from the gear teeth, with fault levels set at 30 %, 70 %, and 100 %. These levels represent partial tooth damage, substantial degradation, and total failure, respectively, as shown in Fig. 4(b). These experiments are crucial for analysing the impact of different fault levels on the system's performance, enabling better prediction and prevention of failures.

Fig. 4(c) illustrates the various transfer pathways of gearbox vibration signals. The second and third pathways influence the signal due to the longer and more complicated propagation via the gearbox, resulting in low sensor-perceived amplitude. In comparison, the first path carries more information about the gearbox condition because it is shorter than the others. Also, the MEMS sensor is orientated along the Z-axis, and the mounting surface is clean and levelled; therefore, the accelerometer is fixed using Loctite 454 adhesive on the first path on the upper side of the gearbox casing. The used adhesive is an insulator and is easily removable using acetone.

Table 4

Statistical features used for ML model.

Peakvalue = $\max x_i $	Meanvalue ($\bar{x}$) = $\frac{1}{N} \sum_{i=1}^N x_i$
Rootmeansquarevalue (x_{rms}) = $\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i)^2}$	Skewness = $\frac{\sum_{i=1}^N (x_i - \bar{x})^3}{(N-1) * \sigma^3}$
Crestfactor (CF) = $\frac{x_{peak-peak}}{x_{rms}}$	Shapefactor = $\frac{\sqrt{\frac{1}{N} \sum_{i=1}^N (x_i)^2}}{\frac{1}{N} \sum_{i=1}^N x_i }$
Standarddeviation (σ) = $\sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$	Kurtosis = $\frac{N \sum_{i=1}^N (x_i - \bar{x})^4}{\sum_{i=1}^N [(x_i - \bar{x})^2]^2}$
Center Frequency (f_{FC}) = $\sum_{i=2}^N \dot{x}_i \dot{x}_i / (2\pi \sum_{i=2}^N x_i^2)$	Root Mean Square Frequency (f_{RMSF}) = $\sqrt{\sum_{i=2}^N \dot{x}_i^2 / (4\pi^2 \sum_{i=2}^N x_i^2)}$
Root Variance Frequency (f_{RVF}) = $\sqrt{f_{RMSF}^2 - f_{FC}^2}$	Mean Frequency (f_{MF}) = $\left[\sum_{i=2}^N \dot{x}(N) \right] N$
Variance (σ^2) = $\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$	

3.4. Feature extraction and selection

Time and frequency domain analysis approaches are the two primary signal analysis methods that researchers widely adopt to evaluate the system's current state. In the time domain analysis, the vibrational signals from the bearing and gear are plotted to time, providing insights into the strength of the vibration signal. Extracting statistical parameters (SPs) from acquired vibration data is a simple and effective approach for time-domain analysis. Root means square value (x_{rms}), crest factor (CF), standard deviation (σ), kurtosis value, skewness, shape factor (SF), peak value, mean value ($\bar{x}$), and variance (σ^2) are some commonly used statistical parameters (SPs) presented in Table 4 [3,4,35]. where, N is the total number of data points, x_i acceleration value for the i^{th} point and $\bar{x}$ is the mean value. Time-domain features like RMS, standard deviation, and peak value provide insights into the overall energy, variability, and extreme changes in vibration signals, which are essential for identifying general changes and potential faults in machinery. Features such as kurtosis, skewness, and crest factor highlight the presence of spikes, asymmetry, and high-amplitude peaks, aiding in the detection of transient faults and irregularities in mechanical operations. The frequency-domain features can capture information that the time-domain features cannot detect.

After conducting an FFT on the obtained vibration signals, several frequency domain features are retrieved from the frequency spectrum. These features include centre frequency (CF), root mean square frequency, mean frequency, and root variance frequency described in Table 4 [36,37]. Frequency-domain features provide vital insights into machinery conditions by analysing the vibration signal's frequency content. Centre frequency identifies the dominant frequency components and any shifts in dynamic behaviour, such as changes in resonant frequencies or operating speeds. Root mean square frequency reflects frequency energy distribution, which is useful for detecting changes due to faults like bearing defects. Root variance frequency measures the spread of frequency components around the RMS frequency, indicating the signal's bandwidth and helping to identify complex vibration patterns or changes in operating conditions. Mean frequency is the weighted average of frequency components, offering insights into the central tendency of the frequency distribution and diagnosing shifts in operating conditions or fault development. Together, these features enable effective fault detection and maintenance planning by providing a comprehensive view of the machinery's frequency characteristics. These metrics are essential for diagnosing the condition of machinery, as they provide insights into the vibration characteristics and potential

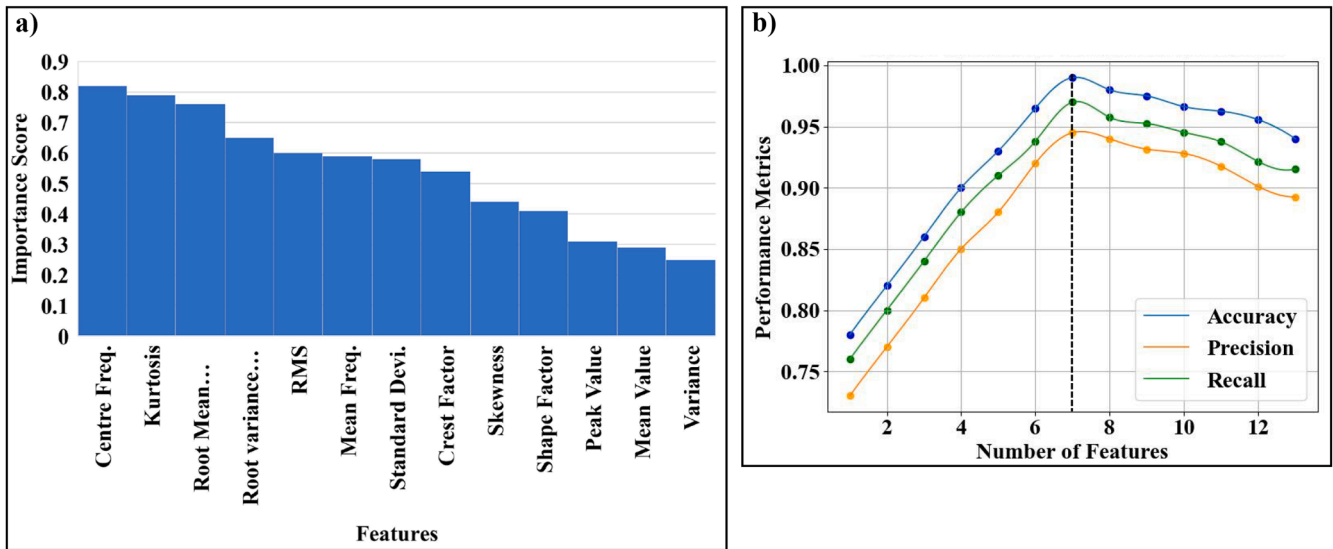


Fig. 5. (a) Feature importance distribution (b) Effect of the number of features on performance metrics.

faults within the system. These selected features are conventional and frequently utilised in previously published articles.

The Random Forest (RF) model for fault classification leverages the Mean Decrease Accuracy (MDA) method to evaluate each vibrational feature's importance, determining its impact on predictive accuracy. MDA begins by establishing a baseline accuracy with all features, then shuffles each feature's values independently, measuring the decrease in accuracy to gauge each feature's significance. A larger drop in accuracy indicates higher importance, while a smaller drop suggests lower relevance. Repeating this process across multiple shuffles produces a stable ranking, allowing RF to retain only the most crucial features. Feature subsets are further assessed by testing the impact on key performance metrics such as accuracy, precision, and recall. Accuracy reflects overall prediction correctness, while precision and recall provide deeper insight into fault detection quality. As the number of features increases, these metrics typically improve until a certain point, after which adding more features can lead to overfitting and a plateau or decline in performance. Fig. 5(a) displays the importance distribution across 13 features, indicating each attribute's contribution to fault classification. Fig. 5(b) shows how performance metrics vary with the number of features, identifying the top seven as the optimal subset. This subset maximises model efficiency and accuracy, focusing only on the most relevant vibrational characteristics. Using the MDA-based selection approach, the RF model achieves high accuracy and robustness, which is crucial for reliable fault detection and classification in mechanical systems.

4. Experimental outcomes

This section describes the comprehensive analysis of experimental results and their implications on the system performance.

4.1. Results of the random forest model

The following section presents the RF model's classification results across various fault conditions and severities.

4.1.1. Functioning of random forest model

The Random Forest (RF) model is a type of ensemble learning that integrates the predictions from several decision trees to improve robustness and performance. RF model reduces the risk of overfitting and increases the accuracy of classification and regression tasks by averaging the outputs of individual trees. The core idea behind the RF model is to introduce diversity into the decision trees by using different

subsets of data and features for each tree, which ensures that the ensemble is not overly dependent on a specific tree and can generalise better to unseen data. Using bootstrap sampling, each tree is trained independently on a randomly selected subset of the training data. At each node, a random subset of features is considered to determine the optimal split. This randomness helps to diversify the trees, making the model more resistant to noise and improving performance. Once the forest of decision trees is built, the RF model aggregates the predictions of all the trees to arrive at a final prediction. In classification tasks, the final class is determined through majority voting, where each tree votes for its predicted class, and the class with the highest votes is selected as the final result. For regression tasks, the final prediction is calculated by averaging the continuous values predicted by all the trees. This aggregation process ensures that the RF model makes more stable and reliable predictions than a single decision tree, particularly in complex datasets with noisy or imbalanced data. Overall, the RF model's ability to generalise well to new data, combined with its robustness to overfitting, makes it one of the most widely used algorithms in machine learning [7, 14].

The Random Forest model operates through several key processes that contribute to its robustness and accuracy. RF model employs bootstrap sampling, where for every tree in the model forest, a unique subsection of data is created by randomly sampling N data points with replacement from the training set. This ensures diversity among the trees because every tree is trained using a distinct subset of the features. In this model, the random feature selection is used at each decision node in a tree. Instead of considering all features, Only a randomly chosen subset of features is considered during the evaluation process, and the best split is chosen from this subset. The random selection of features helps decrease the correlation between individual trees, improving the performance and reducing the risk of overfitting. In the next step, each tree is grown to its full depth without pruning during decision tree construction. For classification tasks, splits are chosen in such a way as to minimise measures like Gini impurity or entropy to improve the tree's ability to separate different classes. For regression tasks, splits are determined to minimise the Mean Squared Error (MSE), which improves the accuracy of predicting continuous values. The model ultimately generates its final prediction by aggregating the outputs from all the trees in the forest. Classification tasks are done through majority voting, where the class that appears most frequently across all trees is selected as the final output. In regression tasks, the average of all the individual tree's results anticipates the final prediction, which leads to a more accurate estimate of continuous values. These combined steps make the

Random Forest model highly effective in classification and regression tasks [38,39].

$$Gini(t) = 1 - \sum_{k=1}^K p_k^2 \quad (4)$$

$Gini(t)$ is the Gini impurity for classification described in Eq. (4), where p_k is the proportion for class k in node t , and K is the total number of classes. This equation measures the purity of a node. A lower Gini impurity indicates a more homogeneous node.

$$Entropy(t) = - \sum_{k=1}^K p_k \log(p_k) \quad (5)$$

$Entropy(t)$ is the entropy for classification described in Eq. (5), where p_k is the probability of class k at node t . Entropy measures the level of disorder or impurity in the node, with lower entropy indicating purer class distribution.

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \bar{y}_i)^2 \quad (6)$$

The Mean Squared Error (MSE) for regression is described in Eq. (6), where y_i is the actual value and $\bar{y}_i$ is the predicted value. MSE quantifies the average of the squared differences between actual and predicted values for regression tasks.

$$\bar{y}_{final} = Mode(\bar{y}_{i,1}, \bar{y}_{i,2}, \dots, \bar{y}_{i,T}) \quad (7)$$

$\bar{y}_{final}$ is the final prediction for classification based on the majority votes described in Eq. (7). Where T is the number of trees and $\bar{y}_{i,t}$ is the predicted class from the tree t for input x_i .

$$\bar{y}_{final} = \frac{1}{T} \sum_{t=1}^T \bar{y}_{i,t} \quad (8)$$

$\bar{y}_{final}$ is the final regression prediction based on averaging described in Eq. (8). Where T is the number of trees and $\bar{y}_{i,t}$ is the predicted value from the tree t for input x_i .

In this process, each decision tree is constructed using random subsets of the acquired data, which helps enhance the model's accuracy while minimising the risk of overfitting. For bearing and gear fault classification, the RF model analyses various features of vibration signals or other sensor data collected from the gearbox. The splitting of decision nodes in each tree is optimised using metrics like gini impurity or entropy, which ensures a clear separation between different fault types, such as healthy or faulty configurations. Once all trees have made their predictions, the RF model aggregates these outputs through majority voting to decide the final classification. This approach makes RF highly effective in identifying and classifying a wide range of bearing and gear faults, offering robustness, accuracy, and resilience against noise in the data.

4.1.2. Outcomes of RF model

In this study, Random Forest (RF) classifiers are employed to detect and classify various gearbox conditions such as healthy, faulty gear, and bearing inner and outer raceway fault. The dataset for training and testing is generated from extracted vibration features. The dataset is divided into 70 % for training and 30 % for testing to assess the model's performance. The training and testing processes were conducted using a Raspberry Pi 4B+ CPU. ML models typically consist of two parameters: model parameters and hyperparameters. Model parameters are automatically learned from the data during training, whereas hyperparameters must be manually adjusted to optimise the model's performance. Hyperparameter tuning is essential for achieving high

Table 5

Performance measures expressions.

Performance measures	Healthy	BIRF	BORF	GF
$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$	99.56 %	99.4 %	99.66 %	99.47 %
$Recall = \frac{TP}{TP + FN}$	98.75 %	99.2 %	99.5 %	98.9 %
$Precision = \frac{TP}{TP + FP}$	99.49 %	98.7 %	99.15 %	98.99 %
$F1 - Score = \frac{TP}{TP + \frac{1}{2}(FP + FN)}$	99.12 %	98.95 %	99.32 %	98.94 %

accuracy and minimising errors in machine learning models. In the case of the Random Forest (RF) model, hyperparameters such as the number of iterations, the splitting criterion, and the number of features selected for the best-split log2 were optimised. The entropy equation was selected as the splitting criterion to minimise classification errors by ensuring purer node splits. The optimal number of iterations was 800 through grid search optimisation, achieving a balance between computational efficiency and accuracy. This tuning process ensures the RF model delivers robust and accurate classification results. The performances of the ML model are evaluated according to Table 5.

The classification outcomes from the RF machine learning model are depicted in Fig. 6. The dataset's relevant confusion matrices show that the classifier excels at identifying bearing and gear faults. As shown in Fig. 6(a), the machine learning algorithm achieved an overall accuracy of 99.09 %. The effectiveness of the RF model is further assessed through the receiver operating characteristic (ROC) curve and the area under the curve (AUC), as depicted in Fig. 6(b). The ROC measures the model's effectiveness, with higher AUC values indicating a more accurate classification model that better differentiates between system states. The ROC curves for the RF model represent the ability to differentiate between four classes (1-healthy, 2-inner raceway fault, 3-outer raceway fault, and 4-gear fault) are presented in Fig. 6(b). These results confirm that the proposed RF model effectively distinguishes between the different classes, achieving AUC values of 0.96, 0.98, 0.96, and 0.96 for classes 1, 2, 3, and 4, respectively.

Additionally, micro-average and macro-average in ROC analysis are employed to determine the model's performance in an imbalanced dataset. The macro-average encourages models to focus on every class correctly, whereas the micro-average relies heavily on the majority classes. The AUC details demonstrate that the macro and micro-average values are identical, indicating balanced data across the four classes. In conclusion, the proposed RF method is robust and stable for fault classification. Various performance metrics are used for evaluation, including recall, accuracy, F1-score, and precision.

An accuracy metric assessed the error rate of the model's accurate and inaccurate predictions. The percentage of accurately identified positive samples compared to the total number of positive samples is known as recall. Precision is the percentage of accurately detected positive samples among all anticipated positive samples. Precision and recall are combined into a metric called the F1-score, ranging from 0.0 to 1.0, where higher values signify better overall model performance [40]. Additionally, the train-test split method was employed to ensure that the datasets used for training and testing were representative of the entire dataset. A 70/30 split was chosen, balancing sufficient data for model training and a reliable dataset for performance determination. The 70 % training set offers sample data for the model to learn effectively, while the 30 % test set is sufficiently large to estimate the model's performance accurately.

The expressions of accuracy, precision, recall, and F1-score are summarised in Table 5. The developed ML model demonstrated robust results, achieving high accuracy, precision, recall, and F1 scores across all fault classes. These metrics indicate the model's effectiveness in correctly classifying different types of faults, thereby validating the reliability of the MEMS setup for fault detection in rotating machinery.

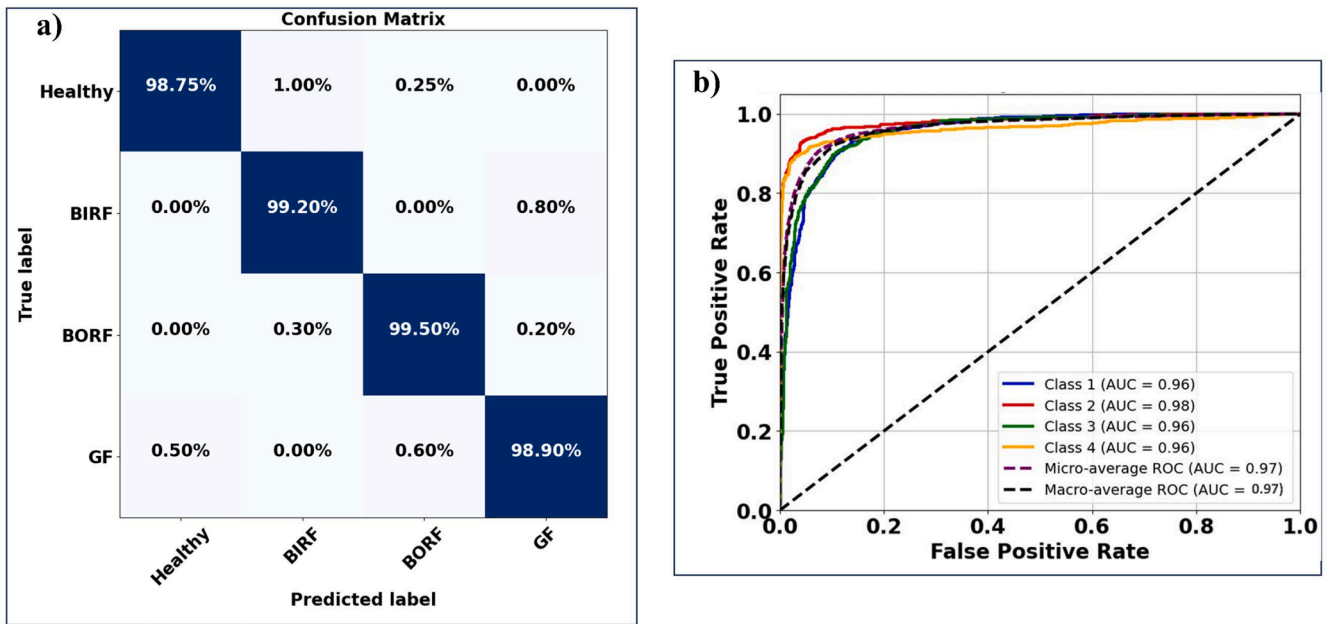


Fig. 6. (a) Confusion matrix (b) Receiver operating characteristic (ROC) curve.

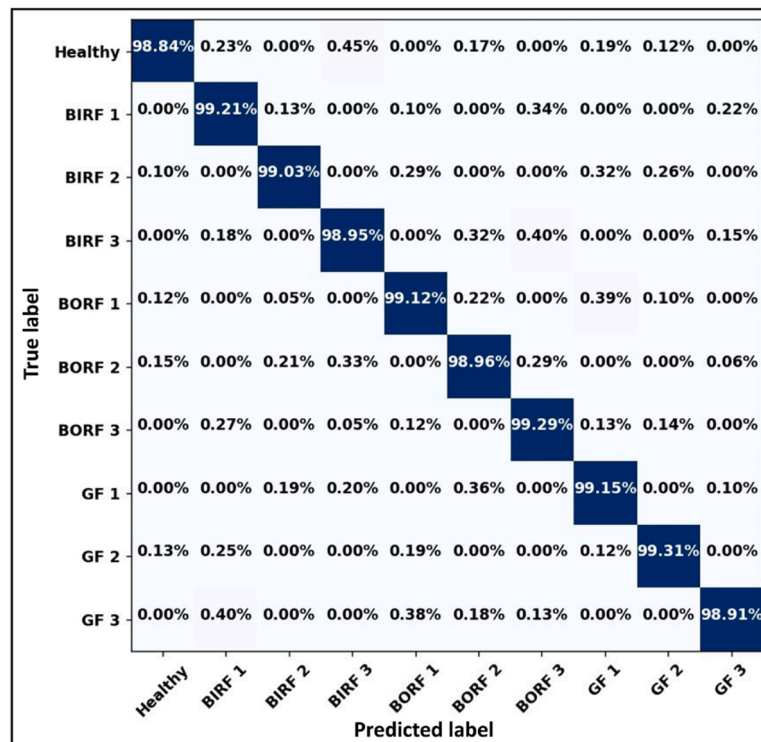


Fig. 7. Confusion matrix for classifying faults with varying fault severities.

The high precision values also confirm the setup's capability to minimise false positives, ensuring accurate fault identification. The successful evaluation of these performance metrics underscores the potential of the MEMS setup for condition monitoring and maintenance applications. These performance metrics have significant practical implications for real-world fault detection and maintenance practices. High accuracy ensures reliable fault detection, minimising false alarms and unnecessary maintenance actions. High precision and recall indicate that the model effectively identifies actual faults without missing significant issues, allowing for timely interventions. The F1 score ensures a balanced

performance, which is critical for maintaining the reliability and efficiency of machinery.

Fig. 7 describes the confusion matrix for classifying faults with varying fault severities. The developed machine learning model estimated fault types and severities by distinguishing between healthy and fault conditions. In the confusion matrix, BIRF 1 (minor degradation), BIRF 2 (moderate degradation), and BIRF 3 (severe failure) represent bearing inner raceway faults severity levels. Similarly, BORF 1 (minor degradation), BORF 2 (moderate degradation), and BORF 3 (severe failure) correspond to bearing outer raceway faults severity levels. GF 1,

Table 6
Comparative studies for fault classification.

Ref.	Feature Extraction	Modelling Technique	Fault Types Examined	Performance Results	Application and Complexity	Key Findings
This research article	Time-domain and frequency-domain vibrational features. Mean Decrease Accuracy (MDA) is used for feature selection.	MDA and RF model for feature selection and classification.	Gearbox faults across the inner ring, outer ring, and gear tooth vary with fault size at different RPM and load conditions.	Precision: 97 % Recall: 95 % Overall Accuracy: 96 % To classify faults. 99.077 % overall accuracy in classifying faults with varying fault severities	Designed for MEMS-based, real-time industrial applications with high precision.	RF model prioritises significant vibrational features to effectively classify faults, with MEMS capability for detailed system monitoring.
[41]	FFT, combined with temperature analysis for noise reduction, optimised with ensemble algorithms for early detection.	Ensemble RF, XGBoost (XGB), Light XGB, and Logistic Regression models.	Gearbox failure, bearing failure, and normal operating conditions in wind turbine generators (WTG).	Accuracy: 99.994 %, Recall, 99.982 %, F1 Score, 98.124 %, Kappa, 99.330 %.	Complex, multi-stage model for early drivetrain and bearing fault detection in wind turbines	The ensemble model outperforms traditional methods, detecting faults 5 days early with high accuracy, validated by finite element modelling, Boroscope, and vibration analysis.
[42]	Texture features are extracted using Local Binary Pattern (LBP) on grayscale images of vibration signals.	Multiple models (K-NN, ANN, Naive Bayes) are tested, with ANN achieving the highest classification accuracy.	Bearing faults across the inner ring, outer ring, and ball, with fault size variations and different RPMs.	Accuracy: 100 % for fault type and size detection with ANN A confusion matrix and other performance measures are not included in the study.	The texture-based image approach is suitable for real-time diagnostics in bearing fault detection.	Texture-based LBP features provide high stability and accuracy for fault differentiation, with lower complexity for implementation.
[14]	Time-domain features (kurtosis, skewness, mean, RMS) and Discrete Wavelet Transform (DWT)-based features.	Support Vector Machine (SVM) and RF.	Ball bearing faults	The efficiency of RF is 98.38 % with the ReliefF ranking method.	Application in fault diagnosis of ball bearings, using vibration data from Case Western Reserve University bearing dataset.	Relieff ranking improves feature selection and accuracy, with RF outperforming other combinations and achieving 98.39 % cross-validation accuracy.
[43]	A one-dimensional local binary pattern (1D-LBP) approach extracts the time domain features.	Gray relational analysis (GRA), KNN, ANN, SVM and LDA.	NS (Normal State), IRF (Inner Raceway Fault), ORF (Outer Raceway Fault), and BF (Ball Fault).	100 % accuracy for NS, ORF, BF, IRF, and fault size. 99.044 % and 99.224 % at speed-300 and 60 rpm intervals.	Designed for laboratory setups; future applications planned for real-time data in operational settings.	The proposed model classifies bearing fault types, sizes, and rotational speed magnitudes.
[44]	Time-domain and frequency-domain vibrational features.	OAA-MCSVM, RF, and Fk-NN.	Bearing faults, Rotor faults, and Mass unbalance.	The OAA MCSVM, RF, and Fk-NN ML models acquire 100 %, 86 %, and 80 % accuracy, respectively.	Cost-effective machinery fault diagnosis using MEMS-based sensors.	MEMS-based sensors provide comparable reliability to conventional sensors, with high accuracy in identifying motor faults.
[45]	One-dimensional ternary patterns (1D-TP) for feature extraction.	RF, SVM, ANN, k-NN and BayesNet.	NS (Normal State), IRF (Inner Raceway Fault), ORF (Outer Raceway Fault), and BF (Ball Fault).	100 % accuracy for NS, ORF, BF, IRF and fault size. 91.25 % for different speed datasets.	Suitable for detecting fault size and location in bearings.	The proposed model Achieved high accuracy across fault types, sizes, and speeds.
[46]	Random multi-grained scanning and PCA feature extraction.	Random multi-grained fusion cascade forest (rgfc-Forest).	Bearing ball, inner and outer raceway fault and air gap eccentricity.	84.41 % for motor System Diagnosis and 92.72 % for the gearbox system. The precision, Recall, and ROC curves show superior results.	Effective for fault diagnosis in electro-mechanical systems with small datasets.	Achieves high accuracy with limited samples, outperforming traditional deep learning and deep forest models in feature representation and classification.
[47]	Maximal Overlap Discrete Wavelet Packet Transform (MODWPPT), Spectral Proper Orthogonal Decomposition (SPOD) and Time-domain features (DAMV, ASM, EWL).	RF, Ensemble Tree, KNN and MSVM.	Artificial and real inner and outer ring defects and external ball bearing faults.	The Proposed approach is more efficient for bearing state detection under different operation modes.	Used for external ball bearing fault diagnosis, reducing hardware needs by applying motor current and vibration signals.	MCSA effectively diagnoses external bearing faults in electro-mechanical systems, outperforming Accugram with slightly lower accuracy.

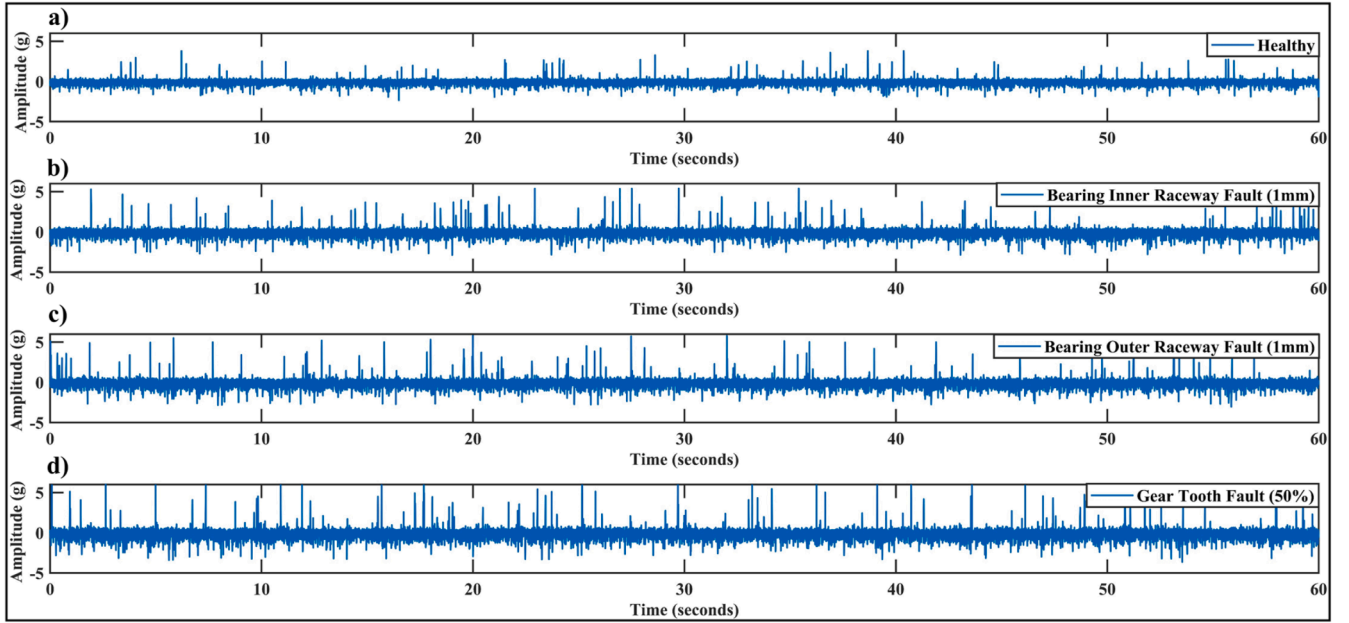


Fig. 8. Time domain plots at 30 Hz input speed under variable loading conditions for (a) Healthy configuration, (b) Bearing inner raceway 1 mm fault, (c) Bearing outer raceway 1 mm fault and (d) Gear tooth fault (50 % tooth damage).

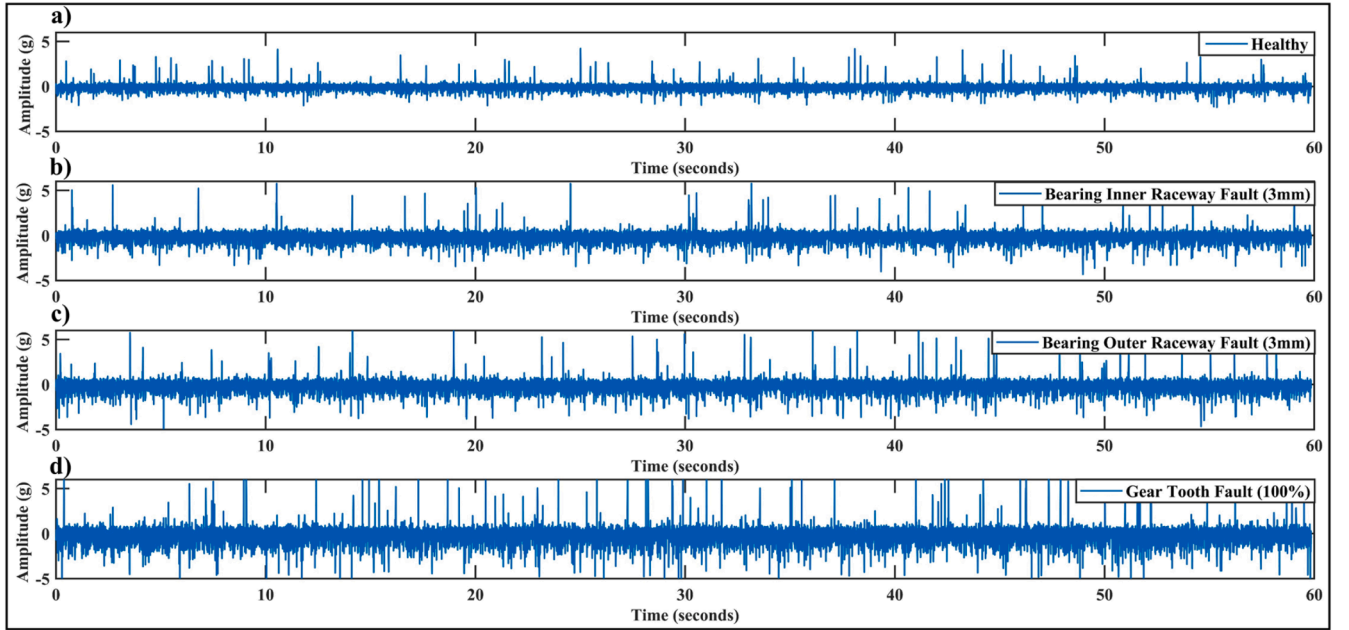


Fig. 9. Time domain plots at 50 Hz input speed under variable loading conditions for (a) Healthy configuration, (b) Bearing inner raceway 3 mm fault, (c) Bearing outer raceway 3 mm fault and (d) Gear tooth fault (100 % tooth damage).

GF 2, and GF 3 represent partial tooth damage, substantial degradation, and total failure of gear tooth respectively. The model's overall accuracy in classifying fault severity is 99.077 %, demonstrating its high performance in distinguishing between various fault types and severities. This categorisation provides a detailed fault severity assessment, enhancing the model's diagnostic precision.

4.1.3. Comparative analysis with the existing literature

Table 6 provides a detailed comparison of bearing and gear fault classification methods reported in the literature. Previous studies have reported success rates and accuracy ranging from 91 % to 100 %. In this study, the RF ML model is developed to classify different types of faults

based on features extracted from the vibration data. Fig. 6 displays the model's overall accuracy of 99.09 % in classifying different types faults. Similarly, Fig. 7 describes the overall accuracy for classifying faults with different severity levels is 99.077 %. These results highlight the model's ability to evaluate the precision and reliability of the MEMS setup in accurately capturing and classifying vibration signals. Key performance metrics such as accuracy, precision, recall, and F1 score are analysed to validate the effectiveness of the developed MEMS setup in detecting various faults in the gearbox.

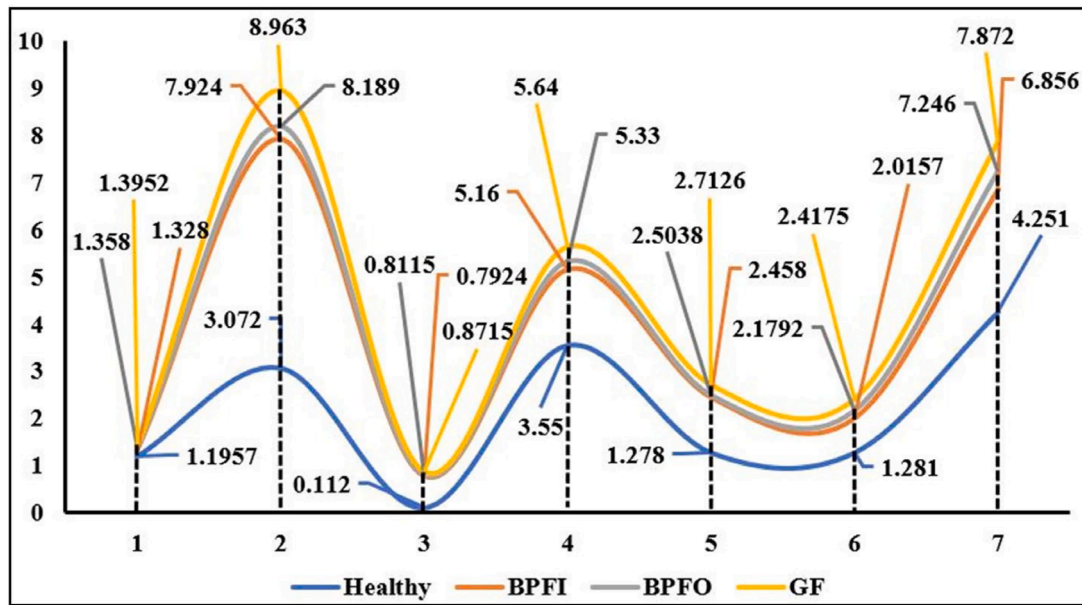


Fig. 10. Extracted time domain statistical at 50 Hz input speed under variable loading conditions for healthy, bearing inner raceway, bearing outer raceway and faulty gear configurations.

4.2. Time domain statistical parameters extraction and analysis

Extracting statistical parameters (SPs) from acquired vibration data is a simple and effective approach for time-domain analysis, as described in Section 3.4. Fig. 8 (a, b, c & d) shows the time domain responses at 30 Hz input constant speed under variable loading conditions for healthy, bearing inner raceway fault (1 mm hole at inner raceway), bearing outer raceway fault (1 mm hole at outer raceway), and faulty gear configurations (50 % tooth damage). The plots reveal that the amplitude of the vibrations is comparatively higher in faulty configurations than in healthy ones. The peak values of vibration amplitude are 3.851 g and 5.856 g, 6.2458 g & 7.272 g during healthy, bearing inner raceway fault, bearing outer raceway fault and faulty gear configurations, respectively. This increased amplitude of vibrations during faulty configurations results from the bearing and gear fault, signifying an alteration in all vibration levels.

Similarly, Fig. 9 (a, b, c & d) represents time-domain responses at 50 Hz input rpm under variable loads for healthy, inner raceway fault (3 mm hole), outer raceway fault (3 mm hole), and faulty gear (100 % tooth damage) configurations. Vibration amplitudes are higher in faulty cases, with peak values of 4.251 g, 6.156 g, 6.446 g, and 7.872 g for healthy, inner, outer, and gear faults, respectively. The results reveal that the vibration intensity increases significantly as fault severity and input rpm rise, highlighting fault signatures and enhancing the distinction between healthy and faulty states.

Statistical parameters (SPs) are extracted from the acquired time-domain vibration data to assess vibration signal strength for fault detection. These SPs play a crucial role in identifying potential faults within the system. As illustrated in Fig. 10, the SPs from 1 to 7 describe the Rms, kurtosis, standard deviation, crest factor, skewness, shape factor and peak value, respectively. A sudden increase in the values of all SPs is observed in the faulty cases compared to the healthy configurations. This noticeable spike in SPs serves as an initial indication of a fault in the system. The increased values of the statistical parameters result from the bearing and gear fault, signifying an alteration in the overall signal strength. The time domain analysis is a preliminary analysis to observe the signal strength. Still, this analysis does not provide information about the reciprocal impact of bearing faults on gear vibration signatures and vice versa. To achieve this, identifying characteristic frequencies of gearbox components is crucial, and this is carried out

Table 7

Characteristic frequencies.

Characteristic frequencies of bearing and meshed gears	
$bpfo = rpm \cdot \frac{N_B}{2} \left(1 - \frac{B_d}{P_d} \cos(\alpha) \right) =$	$gmf = f_p * T_p(Hz), f_g * T_g(Hz) =$
251.66 Hz	544 Hz

using the envelope spectrum approach. Envelope analysis is one of the commonly used approaches for vibration signal analysis based on characteristic frequency, which gives information related to the location of fault by recognising the fault frequencies.

4.3. Identification of characteristics frequencies using envelope analysis

Envelope analysis of the signals gives a detailed description regarding the location of fault by recognising the characteristic frequencies. This analysis uses the Hilbert Transform (HT) approach for amplitude demodulation. HT is a commonly used method for detecting faults in bearings and gears [48,49]. This technique is focused on extracting modulation side bands from the amplitude spectrum of the vibration signal to identify fault-related frequencies. In the case of bearing outer raceway faults or gear faults, modulation sidebands appear at harmonics of the fundamental fault frequency or gear mesh frequency. The methodology involves acquiring the raw vibration signal, demodulating it to emphasise the envelope, and analysing the resulting spectrum to identify fault-related side bands. This precise identification allows for proactive maintenance measures, preventing catastrophic failures and minimising downtime. In variable loading conditions, envelope analysis integrating amplitude demodulation proves effective by isolating fault-related modulation sidebands from dynamic background noise, ensuring robust fault detection in real-world industrial scenarios where machinery operates under fluctuating loads.

The envelope analysis also helps to detect the high-frequency vibrations caused by sudden failure in mechanical parts of the gearbox. In the bearings, when the balls of the bearing make contact with a faulty outer surface of the bearing, a periodic vibration is induced in the produced vibrational signal. This periodic vibration is analysed using the ball pass frequency outer (bpfo) as described in Table 7. The increased

Table 8
Bearing and gears specifications.

Ball Bearing (6007-2RS)		Gear Specifications	
Shaft rotation frequency (f_r)	50 Hz	Gear type	Helical Gears
N_b (Number of balls in the bearing)	13	T_p (number of teeth on the pinion)	27
P_d (Pitch diameter)	48.75 mm	T_g (number of teeth on the gear)	85
B_d (Diameter of the ball)	11 mm	f_p (rotating frequency of the pinion)	20.13 Hz
α (Contact angle of the ball)	0°	f_g (rotating frequency of the gear)	6.4 Hz
D_o (Diameter of the outer raceway)	62.5 mm	G_r (gear ratio)	3.125
D_i (Diameter of the inner raceway)	35 mm	gmf	544 Hz

amplitude of the harmonics of $bpfo$ provides information about the severity of the fault. The gear mesh frequency (gmf) explained in Table 7 is used to analyse the gear fault in the system. The gmf is related to the number of teeth and the rotational speed of the gears in the mesh. The gmf represents the frequency at which the teeth of two gears engage and disengage during rotation. Gear faults can introduce changes in the vibration signal at the gmf and its harmonics, providing valuable information about the gear condition.

Table 8 shows the complete specification of the bearing used for research, where N_b is the number of balls in the bearing, P_d is the pitch diameter, B_d is the diameter of the ball, α is the contact angle of the ball, D_o is the diameter of the outer raceway, D_i is the diameter of the inner raceway. Specifications of the meshed pair of gears are presented in Table 8, where T_p is the number of teeth on the pinion, T_g is the number of teeth on the gear, f_p is the rotating frequency of the pinion, f_g is the rotating frequency of the gear, and G_r is the gear ratio.

Fig. 11 shows the complete envelope spectrum analysis for the healthy configuration of the system. In the frequency plot analysis, the absence of prominent side bands or unexpected frequency components suggests a lack of significant structural or mechanical issues. The spectral signature is organised and well-defined, indicating steady and healthy operating conditions. The envelope spectrums shown in Fig. 11 (a), (b) & (c) describe the amplitude of f_r , $bpfi$, $bpfo$, and gmf at 49.5 Hz,

250.523 Hz, 399.547 Hz, and 547.236 Hz frequencies, respectively. The insignificant change in the envelope spectrum and missing sidebands across the characteristic frequencies indicate that the system is operating within the expected parameters. This outcome aligns with the anticipated frequency response of a well-maintained and healthy system.

Fig. 12 shows the complete envelope spectrum analysis of the system's faulty gear configuration. As shown in Fig. 12(b) & (c), the elevated amplitudes of gmf and its harmonics at 545.386 Hz and 1091.11 Hz frequencies strongly suggest the presence of a gear fault within the machinery. Table 9 describes the experimentally observed characteristic frequencies such as $bpfo$ (251.727 Hz), $bpfi$ (400.364 Hz), gmf , and sidebands across the first and second harmonics of gmf . The sidebands across the first (545.386 Hz) and second harmonics (1091.11 Hz) of gmf are visible in Fig. 12(b) & (c), and the expressions of harmonics are described in Table 9, giving a strong indication of gear tooth fault. Fig. 12(a) shows a seemingly stable envelope spectrum with no appreciable changes. The absence of sidebands at bearing defect frequencies indicates an ostensibly healthy bearing. However, a crucial observation arises as the amplitudes of $bpfi$ at 251.727 Hz frequency and $bpfo$ at 400.364 Hz frequency exhibit an unexpected increase, as shown in Fig. 12(a). A gear fault causes this increase in amplitude. The gear fault in a gearbox generates interruptions in the rotational motion, causing excessive loads on supported bearings. As a result, the increased amplitude of harmonics of bearing characteristics frequency is visible in the plot, and side bands across the harmonics lead to different patterns in the vibration signature. This anomaly alters the overall spectrum and possibly leads observers to misinterpret a bearing defect. This finding underlines the complex nature of fault identification, underlining the importance of an integrated approach to vibration signature interpretation in the pursuit of more accurate condition monitoring.

Fig. 13 shows the complete envelope spectrum analysis of the system's faulty bearing configuration. As shown in Table 10 and Fig. 13(a), the harmonics of $bpfo$ at 251.273 Hz, 552.915 Hz, 754.57 Hz, and 1005.26 Hz frequencies strongly indicate the presence of bearing outer race fault. This fault causes irregularities in the rotating speed of the shaft; as a result, this variation in speed can impact the gmf , as shown in Fig. 13(b). The amplitude of the gmf at 545.221 Hz frequency is more significant than that of the healthy configuration, and the sidebands are

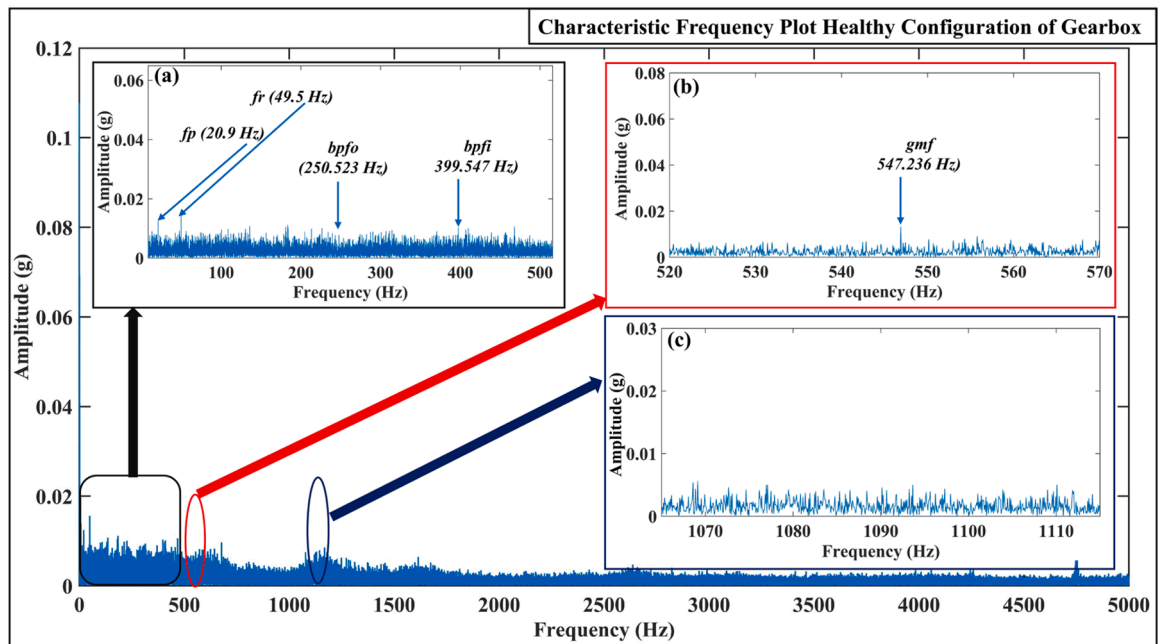


Fig. 11. Envelope Spectrum of Healthy Configuration at 50 Hz input speed under variable loading conditions.

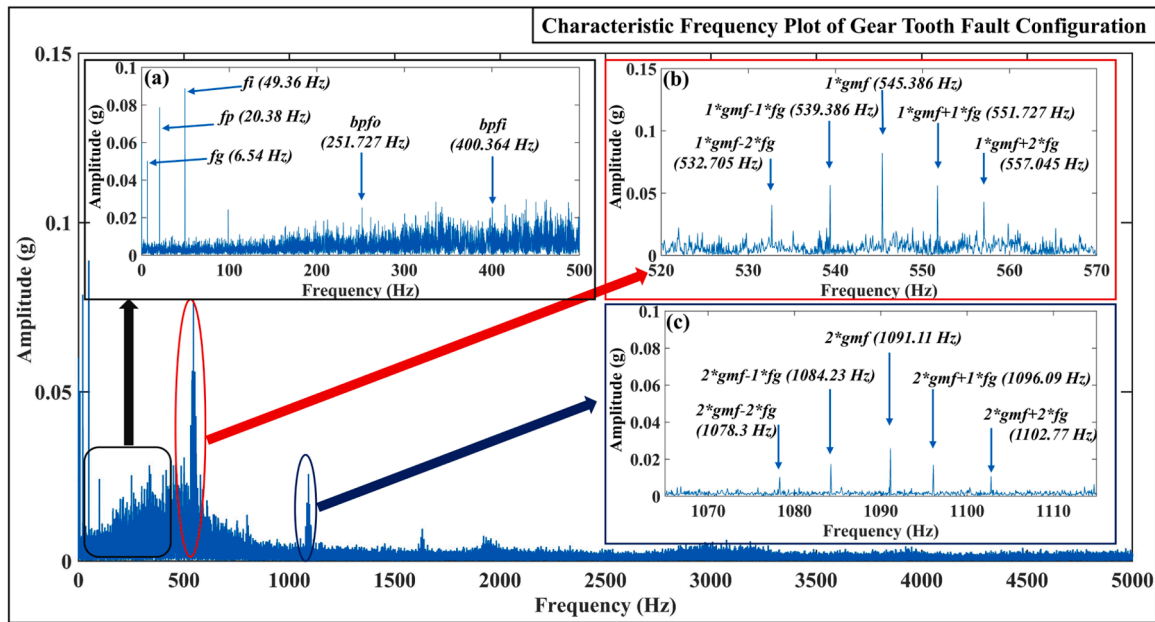


Fig. 12. Envelope Spectrum of Faulty gear Configuration at 50 Hz input speed under variable loading conditions.

Table 9

Experimentally observed the frequency of faulty gear configuration.

Shaft rotation frequency (f_r)	50 Hz	Gear rotation frequency (f_g)	6.54 Hz
Ball pass frequency outer ($bpfo$)	251.727 Hz	Gear Mesh Frequency ($1*gmf$)	545.386 Hz
Ball pass frequency inner ($bphi$)	400.364 Hz	Sidebands across the first harmonics of characteristic frequency (gmf) = $f = m*gmf \pm n*f_g$ (m and n denote the relative integers)	$1*gmf - 2*f_g$ ~ 532.705 Hz $1*gmf - 1*f_g$ ~ 539.386 Hz $1*gmf + 1*f_g$ ~ 551.727 Hz $1*gmf + 2*f_g$ ~ 557.045 Hz
		Gear Mesh Frequency ($2*gmf$)	1091.11 Hz
		Sidebands across the Second harmonics of characteristic frequency $f = m*gmf \pm n*f_g$ (m and n denote the relative integers)	$2*gmf - 2*f_g$ ~ 1078.3 Hz $2*gmf - 1*f_g$ ~ 1084.23 Hz $2*gmf + 1*f_g$ ~ 1096.09 Hz $2*gmf + 2*f_g$ ~ 1102.77 Hz

also apparent. This could potentially confuse the observer regarding the presence of a gear defect. In addition to these observations, the presence of bearing defects introduces further complexities to the vibration signature. Bearing defects contribute additional dynamic forces and vibrations, which can exacerbate the modulation of the gmf .

5. Conclusion

This study investigates the detection and analysis of bearing and gear faults using a MEMS-based vibration measurement system. The developed MEMS setup ensures that the vibration data is captured at the desired sampling frequency and efficiently stored in CPU memory. Further, an RF model is also developed to classify different types of faults based on features extracted from the acquired vibration data. The model evaluates the precision and reliability of the MEMS setup in capturing and classifying vibration signals. A detailed signal analysis is also conducted to evaluate the performance of the developed MEMS setup and

examine the impact of bearing vibration signatures caused by gear faults and vice versa.

The primary outcomes of this research are:

1. The RF model exhibited high accuracy, recall, and F1 scores across all fault classes, demonstrating its strong performance in detecting and classifying faults. The model's robust ROC and AUC metrics further confirm its reliability. These results validate the MEMS setup as a promising condition monitoring and predictive maintenance tool in rotating machinery.
2. Time-domain analysis revealed significant spikes in the signal power (SPs), serving as early indicators of faults. The envelope spectrum analysis confirmed bearing defects, as evidenced by sidebands around the bearing's outer raceway fault frequencies ($bpfo$) and gear mesh frequency (gmf). These further underscores the MEMS setup's efficacy in fault detection.
3. Increased vibration amplitudes at the gear mesh frequency and harmonics, along with the appearance of sidebands, are observed in the presence of bearing faults. Conversely, gear faults led to higher vibration amplitudes at frequencies linked to bearing defects, indicating the interplay between bearing and gear faults.
4. Bearing defects cause irregular gear motion, leading to enhanced modulation of the gear mesh frequency (gmf) and an increased amplitude, which could result in further issues, including potential gear failure.
5. Gear faults cause rotational motion disruptions, imposing additional bearing loads. These disruptions generate altered vibration signatures, characterised by increased harmonics and sidebands in the bearing frequency range.

Future work should improve sensor capabilities, apply the method to more complex systems, and assess its performance in real-time operational environments with varying conditions. The MEMS approach should also be further developed for edge computing and real-time processing within IoT frameworks.

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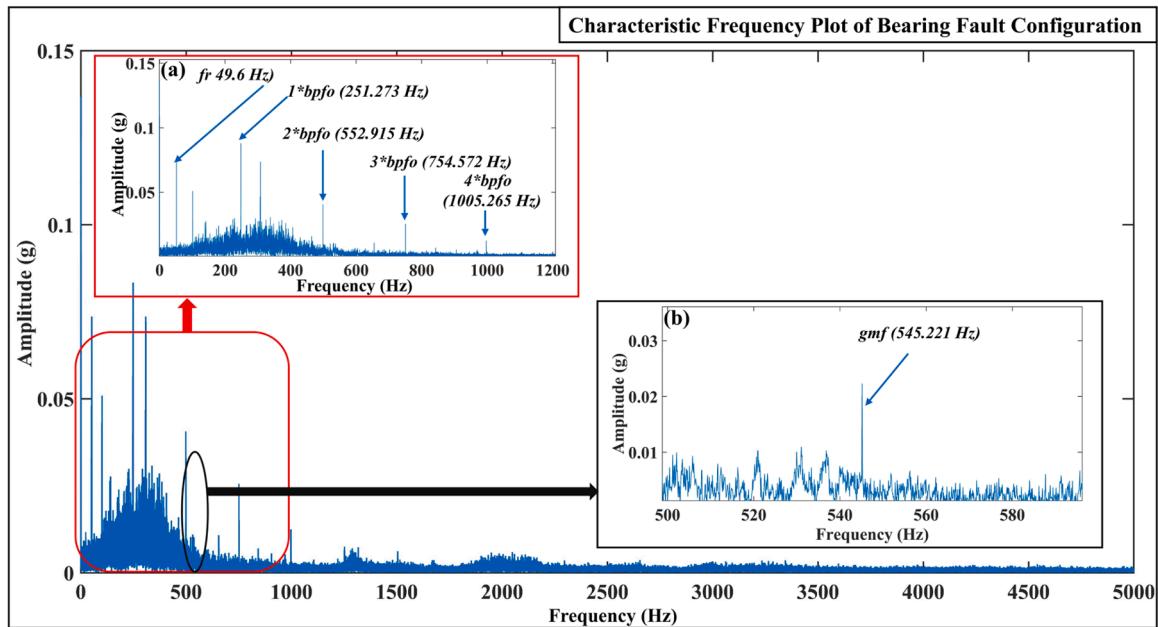


Fig. 13. Envelope Spectrum of Faulty bearing Configuration at 50 Hz input speed under variable loading conditions.

Table 10

Experimentally observed the frequency of faulty bearing configuration.

Shaft rotation frequency (f_r)	50 Hz	Gear rotation frequency (f_g)	6.45 Hz
Ball pass frequency outer (bpfo)	251.273 Hz	Gear mesh frequency (gmf)	545.221 Hz
harmonics of characteristic frequency (bpfo)	1. bpfo ~ 251.273 Hz	Side Bands across the first harmonics of characteristic frequency (gmf) = $f = m \cdot gmf \pm n \cdot f_g$ (m and n denote the relative integers)	Small sidebands are observed across the characteristic's frequency
	2. bpfo ~ 552.915 Hz		
	3. bpfo ~ 754.572 Hz		
	4. bpfo ~ 1005.265 Hz		

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CRediT authorship contribution statement

Gagandeep Sharma: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tejvir Kaur:** Writing – review & editing, Visualization, Validation, Supervision, Project administration. **Sanjay Kumar Mangal:** Validation, Supervision, Resources, Project administration, Conceptualization. **Amit Kohli:** Writing – review & editing, Visualization, Investigation, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that no conflict of interest related to the research article is available. There is no use of AI and AI-assisted technologies in the writing process.

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Data availability

Data will be made available on request.

References

- [1] M. Vasić, B. Stojanović, M. Blagojević, Failure analysis of idler roller bearings in belt conveyors, Eng. Fail. Anal. 117 (2020), <https://doi.org/10.1016/j.engfailanal.2020.104898>.
- [2] M.O. Jakobsen, E.S. Herskind, K. Bjerger, P. Ahrendt, C.F. Pedersen, M.B. Knudsen, Vibration signatures in ball bearings as a function of lubricant viscosity ratio κ , under alternating lubrication conditions, Tribol. Int. 156 (2021), <https://doi.org/10.1016/j.triboint.2020.106840>.
- [3] S. Kumar, D. Goyal, R.K. Dang, S.S. Dhami, B.S. Pabla, Condition based maintenance of bearings and gears for fault detection – a review, Mater. Today Proc. 5 (2018) 6128–6137, <https://doi.org/10.1016/j.matpr.2017.12.219>.
- [4] C. Malla, I. Panigrahi, Review of condition monitoring of rolling element bearing using vibration analysis and other techniques, J. Vib. Eng. Technol. 7 (2019) 407–414, <https://doi.org/10.1007/s42417-019-00119-y>.
- [5] M.K. Liu, M.Q. Tran, P.Y. Weng, Fusion of vibration and current signatures for the fault diagnosis of induction machines, Shock Vib. 2019 (2019), <https://doi.org/10.1155/2019/7176482>.
- [6] A.A. Soomro, M.B. Muhammad, A.A. Mokhtar, M.H. Md Saad, N. Lashari, M. Hussain, U. Sarwar, A.S. Palli, Insights into modern machine learning approaches for bearing fault classification: a systematic literature review, Results. Eng. 23 (2024), <https://doi.org/10.1016/j.rineng.2024.102700>.
- [7] B.S. Yang, X. Di, T. Han, Random forests classifier for machine fault diagnosis, J. Mech. Sci. Technol. 22 (2008) 1716–1725, <https://doi.org/10.1007/s12206-008-0603-6>.
- [8] J. Xie, Z. Li, Z. Zhou, S. Liu, A novel bearing fault classification method based on XGBoost: the fusion of deep learning-based features and empirical features, IEEE Trans. Instrum. Meas. 70 (2021) 1–9, <https://doi.org/10.1109/TIM.2020.3042315>.
- [9] R.N. Toma, J.M. Kim, Article bearing fault classification of induction motors using discrete wavelet transform and ensemble machine learning algorithms, Appl. Sci. (Switzerland) 10 (2020), <https://doi.org/10.3390/AP10155251>.
- [10] Y. Kaya, M. Kuncan, E. Acan, K. Kaplan, An efficient approach based on a novel 1D-LBP for the detection of bearing failures with a hybrid deep learning method, Appl. Soft. Comput. 155 (2024), <https://doi.org/10.1016/j.asoc.2024.111438>.
- [11] B.U. Deveci, M. Celtikoglu, O. Albayrak, P. Unal, P. Kirci, Transfer learning enabled bearing fault detection methods based on image representations of single-dimensional signals, Inf. Syst. Front. 26 (2024) 1345–1397, <https://doi.org/10.1007/s10796-023-10371-z>.
- [12] Y. Kaya, F. Kuncan, H.M. Ertunc, A new automatic bearing fault size diagnosis using time-frequency images of CWT and deep transfer learning methods, Turk. J. Electr. Eng. Comput. Sci. 30 (2022) 1851–1867, <https://doi.org/10.55730/1300-0632.3909>.
- [13] E. Acan, M. Kuncan, K. Kaplan, Y. Kaya, Diagnosing bearing fault location, size, and rotational speed with entropy variables using extreme learning machine, J. Braz. Soc. Mech. Sci. Eng. 46 (2024), <https://doi.org/10.1007/s40430-023-04567-2>.

- [14] V. Vakharia, V.K. Gupta, P.K. Kankar, Efficient fault diagnosis of ball bearing using relief and random forest classifier, *J. Braz. Soc. Mech. Sci. Eng.* 39 (2017) 2969–2982, <https://doi.org/10.1007/s40430-017-0717-9>.
- [15] K.R. Williams, R.S. Muller, Etch rates for micromachining processing, *J. Microelectromech. Syst.* 5 (1996) 256–269.
- [16] G.T.A. Kovacs, N.I. Maluf, K.E. Petersen, Bulk micromachining of silicon, *Proc. IEEE* 86 (8) (1998) 1536–1551.
- [17] A. Albarbar, S. Mekid, A. Starr, R. Pietruszkiewicz, Suitability of MEMS accelerometers for condition monitoring: an experimental study, *Sensors* 8 (2008) 784–799, <https://doi.org/10.3390/S8020784>, 2008, Vol. 8, pp. 784–799.
- [18] A.P. Ompusunggu, K. Eryilmaz, K. Janssen, Condition monitoring of critical industrial assets using high performing low-cost MEMS accelerometers, in: *Proceedings of the Procedia CIRP*, Elsevier B.V., 2021, pp. 1389–1394, <https://doi.org/10.1016/j.procir.2021.11.234>.
- [19] A. Bahador, C. Du, H.P. Ng, N.A. Dzulfarnain, C.L. Ho, Cost-effective classification of tool wear with transfer learning based on tool vibration for hard turning processes, *Measurement (Lond)* 201 (2022), <https://doi.org/10.1016/j.measurement.2022.111701>.
- [20] P. Lou, L. Shi, X. Zhang, Z. Xiao, J. Yan, A data-driven adaptive sampling method based on edge computing, *Sensors (Switzerland)* 20 (2020) 1–16, <https://doi.org/10.3390/s20082174>.
- [21] B. Battulga, M.F. Shaikh, J.W. Chun, S.B. Park, S. Shim, S.B. Lee, MEMS accelerometer and hall sensor based identification of electrical and mechanical defects in induction motors and driven systems, *IEEe Sens. J.* (2024), <https://doi.org/10.1109/JSEN.2024.3447869>.
- [22] T.C. Anil Kumar, N. Bhusal Sharma, A.K. Mishra, P. Patil, S. Sarveswara Reddy, R. Bhardwaj, Fault identification model using IIoT for industrial application, *Meas. Sens.* 24 (2022) 100526, <https://doi.org/10.1016/j.measen.2022.100526>.
- [23] O.D. Mohammed, M. Rantatalo, J.O. Aidanpää, U. Kumar, Vibration signal analysis for gear fault diagnosis with various crack progression scenarios, *Mech. Syst. Signal. Process.* 41 (2013) 176–195, <https://doi.org/10.1016/j.ymssp.2013.06.040>.
- [24] A. Kafeel, S. Aziz, M. Awais, M.A. Khan, K. Afaq, S.A. Idris, H. Alshazly, S. M. Mostafa, An expert system for rotating machine fault detection using vibration signal analysis, *Sensors* 21 (2021), <https://doi.org/10.3390/s21227587>.
- [25] L. Bachar, I. Dadon, R. Klein, J. Bortman, The effects of the operating conditions and tooth fault on gear vibration signature, *Mech. Syst. Signal. Process.* 154 (2021), <https://doi.org/10.1016/j.ymssp.2020.107508>.
- [26] S.A. Mohammed, N.M. Ghazaly, J. Abdo, Fault diagnosis of crack on gearbox using vibration-based approaches, *Symmetry (Basel)* 14 (2022), <https://doi.org/10.3390/sym14020417>.
- [27] Z. Feng, M.J. Zuo, Fault diagnosis of planetary gearboxes via torsional vibration signal analysis, *Mech. Syst. Signal. Process.* 36 (2013) 401–421, <https://doi.org/10.1016/j.ymssp.2012.11.004>.
- [28] C.J. Stander, P.S. Heyns, W. Schoombie, Using vibration monitoring for local fault detection on gears operating under fluctuating load conditions, *Mech. Syst. Signal. Process.* 16 (2002) 1005–1024, <https://doi.org/10.1006/mssp.2002.1479>.
- [29] G. Dalpiaz, A. Rivola, R. Rubini, Effectiveness and sensitivity of vibration processing techniques for local fault detection in gears, *Mech. Syst. Signal. Process.* 14 (2000) 387–412, <https://doi.org/10.1006/mssp.1999.1294>.
- [30] E.Y. Kim, A.C.C. Tan, B. Yang, V. Kosse, Experimental study on condition monitoring of low speed bearings: time domain analysis, in: *Proceedings of the 5th Australasian Congress on Applied Mechanics*, Brisbane, Australia, 2007.
- [31] P.A. Delgado-Arredondo, D. Morinigo-Sotelo, R.A. Osorio-Rios, J.G. Avina-Cervantes, H. Rostro-Gonzalez, R.de J. Romero-Troncoso, Methodology for fault detection in induction motors via sound and vibration signals, *Mech. Syst. Signal. Process.* 83 (2017) 568–589, <https://doi.org/10.1016/j.ymssp.2016.06.032>.
- [32] B. Dolenc, P. Boškoski, D. Juričić, Distributed bearing fault diagnosis based on vibration analysis, *Mech. Syst. Signal. Process.* 66–67 (2016) 521–532, <https://doi.org/10.1016/j.ymssp.2015.06.007>.
- [33] S.A. McInerny, Y. Dai, Basic vibration signal processing for bearing fault detection, *IEEe Trans. Ed.* 46 (2003) 149–156, <https://doi.org/10.1109/TE.2002.808234>.
- [34] G.S. Maruthi, V. Hegde, Application of MEMS accelerometer for detection and diagnosis of multiple faults in the roller element bearings of three phase induction motor, *IEEe Sens. J.* 16 (2016) 145–152, <https://doi.org/10.1109/JSEN.2015.2476561>.
- [35] G. Sharma, T. Kaur, S.K. Mangal, Fault diagnosis and prediction with industrial internet of things on bearing and gear assembly, 2022, [10.1504/IJNET.2022.125114](https://doi.org/10.1504/IJNET.2022.125114).
- [36] M.Q. Tran, M. Elsi, K. Mahmoud, M.K. Liu, M. Lehtonen, M.M.F. Darwish, Experimental setup for online fault diagnosis of induction machines via promising IoT and machine learning: towards industry 4.0 empowerment, *IEEe Access.* 9 (2021) 115429–115441, <https://doi.org/10.1109/ACCESS.2021.3105297>.
- [37] A.N. Saberi, A. Belahcen, J. Sobra, T. Vaimann, LightGBM-based fault diagnosis of rotating machinery under changing working conditions using modified recursive feature elimination, *IEEe Access.* 10 (2022) 81910–81925, <https://doi.org/10.1109/ACCESS.2022.3195939>.
- [38] J.R. Quinlan, *Induction of decision trees*, 1986.
- [39] H.N. B. A. Jain, N. Nagaraj, *Permutation decision trees*, (2023). <http://arxiv.org/abs/2306.02617>.
- [40] T. Emmanuel, D. Mpoeleng, T. Maupong, Power plant induced-draft fan fault prediction using machine learning stacking ensemble, *J. Eng. Res. (Kuwait)* 12 (2024) 82–90, <https://doi.org/10.1016/j.jer.2023.10.001>.
- [41] R.M. Arias Velázquez, Bearings faults and limits in wind turbine generators, *Results Eng.* 21 (2024), <https://doi.org/10.1016/j.rineng.2024.101891>.
- [42] K. Kaplan, Y. Kaya, M. Kuncan, M.R. Minaz, H.M. Ertunç, An improved feature extraction method using texture analysis with LBP for bearing fault diagnosis, *Appl. Soft Comput. J.* 87 (2020), <https://doi.org/10.1016/j.asoc.2019.106019>.
- [43] M. Kuncan, An intelligent approach for bearing fault diagnosis: combination of 1D-LBP and GRA, *IEEE Access.* 8 (2020) 137517–137529, <https://doi.org/10.1109/ACCESS.2020.3011980>.
- [44] J.D. Son, B.H. Ahn, J.M. Ha, B.K. Choi, An availability of MEMS-based accelerometers and current sensors in machinery fault diagnosis, *Measurement (Lond)* 94 (2016) 680–691, <https://doi.org/10.1016/j.measurement.2016.08.035>.
- [45] M. Kuncan, K. Kaplan, M.R. Minaz, Y. Kaya, H.M. Ertunç, A novel feature extraction method for bearing fault classification with one dimensional ternary patterns, *ISA Trans.* 100 (2020) 346–357, <https://doi.org/10.1016/j.isatra.2019.11.006>.
- [46] Y. Ming, H. Shao, B. Cai, B. Liu, rgfc-Forest: an enhanced deep forest method towards small-sample fault diagnosis of electro-mechanical system, *Expert. Syst. Appl.* 238 (2024), <https://doi.org/10.1016/j.eswa.2023.122178>.
- [47] A. Afia, F. Gougam, W. Touzout, C. Rahmoune, H. Ouelmokhtar, D. Benazzouz, Spectral proper orthogonal decomposition and machine learning algorithms for bearing fault diagnosis, *J. Braz. Soc. Mech. Sci. Eng.* 45 (2023), <https://doi.org/10.1007/s40430-023-04451-z>.
- [48] Alberto Bellini, Fabio Immovilli, Rubini Riccardo, Diagnosis of bearing faults in induction machines by vibration or current signals: a critical comparison, in: *Proceedings of the IEEE Industry Applications Society. Annual Meeting (43rd : 2008 : Edmonton A)*, 2010, pp. 1350–1359.
- [49] V.N. Patel, N. Tandon, R.K. Pandey, Defect detection in deep groove ball bearing in presence of external vibration using envelope analysis and Duffing oscillator, *Measurement (Lond)* 45 (2012) 960–970, <https://doi.org/10.1016/j.measurement.2012.01.047>.